

North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2019

FINAL REPORT



Prepared for:

North Coast - Skeena First Nations Stewardship Society
Prince Rupert, BC

Prepared by:

David Robichaud¹ and Angela Addison²

¹ **LGL Limited environmental research associates**
9768 Second Street, Sidney, BC, V8L 3Y8

² **North Coast - Skeena First Nations Stewardship Society**
612, 2 Ave W, Prince Rupert, BC, V8J 1H2



EXECUTIVE SUMMARY

Due to the potential impact on multiple transboundary stocks of salmon and groundfish, the North Coast (Areas 3 and 4) recreational fishery must be monitored. Effort and harvest data, as well as biological samples and genetic information, are required for management of fisheries and monitoring of indicator stocks. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) has operated this program since 2015, with support from the Pacific Salmon Commission Northern Fund. The survey consisted of access point interviews and aerial overflights, typically for the months of June through August but also including May in 2019. Lodge-based recreational harvest and effort were monitored by DFO. Survey design and implementation, and analytical methods were similar to and consistent with those from the past several fishing seasons.

From 1 May to 31 August 2019, a total of 2,592 vessel interviews (involving 7,699 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 240 survey shifts. 32 aerial effort surveys were conducted. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, and length measurements from Pacific Halibut (*Hippoglossus stenolepis*). Survey effort was adequate, as relatively good precision was achieved for the harvest estimates for most of the main taxa (all precision goals were met).

Total fishing effort was estimated to be 14,163 vessel trips. In total, an estimated 26,281 Chinook Salmon were caught, with 15,152 fish kept and 11,129 released. It was estimated that 25,926 Coho Salmon were caught, with 23,820 fish kept and 2,107 released. Total Pacific Halibut catch was estimated to be 18,451 individuals, with 10,967 fish kept and 7,484 released. Biomass of harvested Pacific Halibut was estimated at 67.8 net tonnes (149,487 lb), including 33.5 t (73,937 lb) in Area 3 and 34.3 t (75,550 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 89.8 t (197,976 lb) including 44.4 t (97,920 lb) in Area 3 and 45.4 t (100,056 lb) in Area 4. Total harvest estimates included 4,514 Lingcod (*Ophiodon elongates*), 5,787 rockfish (*Sebastes* spp.), and 1,007 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 1,074 Chinook Salmon, collecting length measurements (median 76 cm), adipose fin-clip incidence rates (8% were clipped), and scale samples. For Coho Salmon, 927 fish were biosampled (median fork length 65 cm; 0.7% adipose clipped). Heads were collected from 59 Chinook and 5 Coho salmon for submission to the Salmon Head Recovery Program. Biosampled Pacific Halibut (n=714) had a median length of 82 cm. Average net and round weights were 6.6 kg (14.6 lb) and 8.8 kg (19.3 lb) in Area 3, and 5.8 kg (12.8 lb) and 7.7 kg (17.0 lb) in Area 4, respectively.

During the month of May, fishing effort and Chinook Salmon harvest were estimated at 1,553 boat trips, and 2,234 fish. The Department of Fisheries and Oceans Canada (DFO) expressed interest in a May creel survey because genetic samples would help them resolve stock composition and determine exploitation rates on early-run-timing stocks. During the month of May 2019, we biosampled 269 Chinook Salmon, taking 230 scale samples, and 22 tagged heads.

Recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 20 years, in some cases dramatically. Yet Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. While the increasing trends highlight the need for continued monitoring of this fishery into the future, there is evidence that management measures can keep harvest under control.

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INTRODUCTION

The North Coast recreational fishery (in DFO's Pacific Region, Management Areas 3 and 4) harvests multiple species, mainly targeting Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, but also Pacific Halibut (*Hippoglossus stenolepis*), Lingcod (*Ophiodon elongates*), and other groundfish, rockfish (*Sebastes* spp.), and invertebrate species. Over the past 20 years fishing effort has increased substantially (Robichaud and Addison 2018, 2019) on the North Coast. In light of increasing pressure, and of conservation concerns for some target stocks, there is a need to understand harvest from the Area 3 and 4 recreational fishery, with respect to catch rates and stock origins. And since harvested salmon can originate from BC or US (enhanced and wild) stocks, there is a critical need for this creel program to determine fishery impacts, especially for stocks of concern.

The creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition, and improved scientific understanding of factors that might limit salmon production. In addition to producing harvest and effort estimates with known error, this creel program has the added benefit of collecting direct field observations of catch, which allows for species identifications and harvest quantities to be verified, and biological samples to be collected from the recreational fishery. For example, CWT data, which are important components of the annual stock assessment data, and are required to monitor trends and manage the fisheries for Nass and Skeena river Chinook and Coho salmon stocks, come from the heads that our observers collect from randomly sampled adipose-clipped Chinook and Coho salmon.

The Area 3 and 4 creel survey program has operated during each recreational fishing season since 2008, estimating fishing effort and species-specific catch rates (Van Tongeren and Winther 2010, Robichaud and Addison 2019). Starting in 2015, the responsibility for carrying out the creel survey shifted from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey followed a design similar to that used in the past by Fisheries and Oceans Canada, which, in turn, was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002).

This report documents the 2019 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

SCOPE

The geographic scope of the creel survey was the same as in the previous ten years (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island; Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).

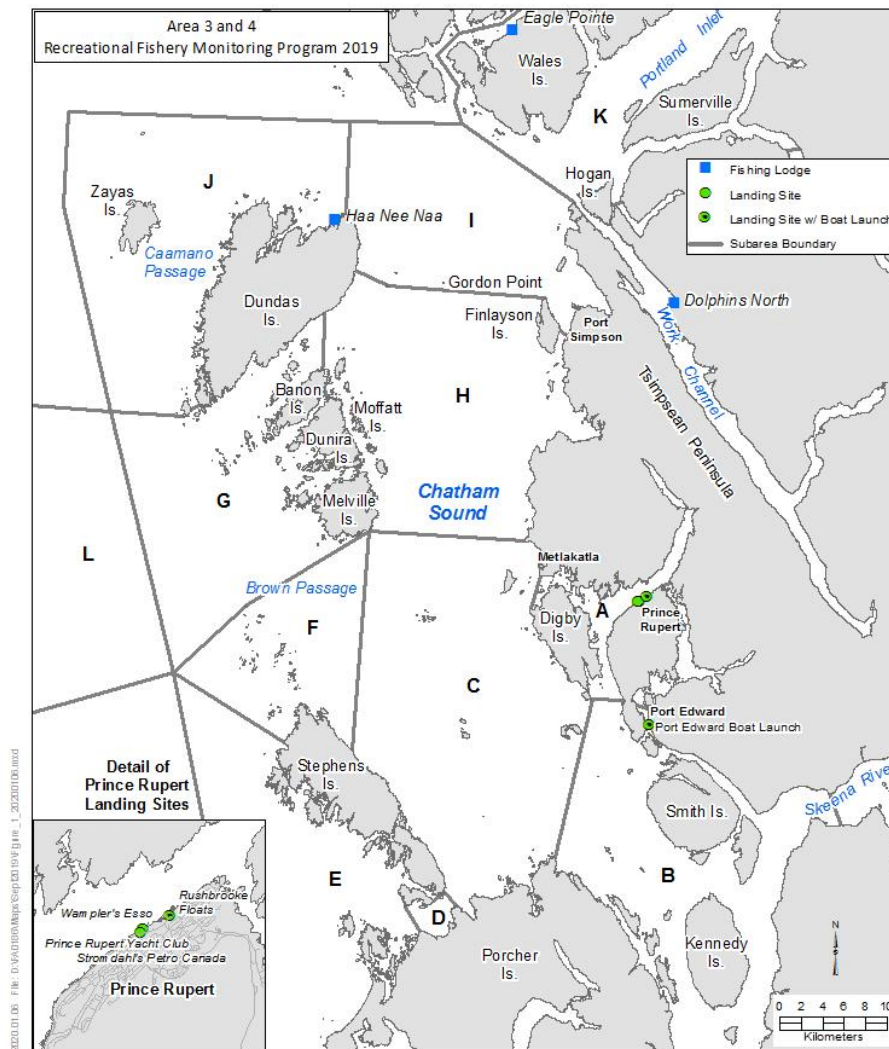


Figure 1. Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas, which were used as spatial strata in the creel survey design.

The temporal scope of the full study was from May 1st through August 31st, 2019. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea.

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut and Lingcod.

GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used recently by DFO (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access

point angler interviews, conducted throughout Areas 3 and 4 from May through August 2019. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month and day type (weekdays vs. weekend days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The seven objectives for this study were:

1. Maintain software for data management and analysis comparable to DFO's CREST program;
2. Provide monthly (May-August) estimates of the catch and effort of all vertebrate species targeted by the recreational fishery and associated estimates of precision;
3. Provide DFO and PSC with monthly in-season catch estimates for all salmon species and Pacific Halibut harvested in the North Coast recreational fishery;
4. Provide communication materials to recreational harvesters to convey the usefulness and importance of their catch data in fisheries management;
5. Collect biological data, such as scale samples and fin clip incidence for Coho and Chinook salmon, and Pacific Halibut lengths from a representative subsample of the landed harvest (during angler interviews);
6. Produce a final written report to be available to a technical audience; and
7. Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests and First Nations.

Objective 1 was addressed over the course of the study period. Objectives 2 and 6 were met by completing this report. Objective 3 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 4 will be met subsequent to the finalization of this report, as its contents will be based on the results presented herein. Objective 5 was met by biosampling fish at access points throughout the study period: fish were measured and sexed, scale samples and fin clip incidence data were collected for Coho and Chinook salmon, heads were collected from adipose-clipped fish, and all biological data were submitted to DFO at the end of study period. Objective 7 was met in December 2019.

METHODS

A complete census of catch and effort was available for lodge-based anglers. However, the remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each month and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1)

conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, harvested fish were measured (nose to fork length for Chinook and Coho salmon; total length for Pacific Halibut) and sexed. Chinook and Coho salmon were checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook and Coho salmon. Chinook Salmon were also examined for flesh colour;
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The analytical methods used, similar to those used in Robichaud et al. (2017) and Robichaud and Addison (2018), provided a statistically unbiased estimate of catch per effort, assuming the anglers interviewed were representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that was conducted from that site in recent years (see Van Tongeren and Winther 2010). Wherever possible, the busiest access points were selected preferentially in order to obtain the maximum number of interviews. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site. The frequency of interview shifts was planned with the target of sampling 20% of estimated effort, while attempting to have “hands on” and direct observations of (and biological samples from) as many fish as possible meant allocating adequate time to each interview. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Ideally, interviews would capture data about angler activity in all creel subareas.

Interviewing sessions were to be separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00 and PM shifts from 13:00 to 21:00. In total, 240 shifts were worked.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 2-3 weekend days and usually on 5-6 weekdays each month. Timing of the overflights was targeted for peak fishing periods, and, although analytical methods allow for the occasional off-peak flight (e.g., due to weather or scheduling conflicts), all flights in 2019 coincided with peak fishing activity.

Angler Activity Patterns

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002). The first weighting factor, $W1$, expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site a , were consistent for landing site a , even during the days when no interviews occurred. A specific $W1$ was calculated for each landing site during each month and day type:

$$W1_{mda} = \frac{N_{md}}{K_{mda}} \quad (\text{Equation 1})$$

where N_{md} was the total number of type d days in month m ; and K_{mda} was the number of days during which interviews occurred at landing site a , on type d days during month m .

The second weighting factor, $W2$, expanded the numbers of interviews conducted, to account for the recreational fishing boats that were *not* interviewed. Some recreational fishing boats were not interviewed because of refusals, and some boats returned during periods of high traffic volume when it was not possible to interview all returning vessels at once. Some returning vessels were not doing recreational fishing at all -- they may have been taking a pleasure cruise, or conducting Food-Social-Ceremonial fishing. For the recreational fishing vessels, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific $W2$ was calculated for each surveyor (u) operating on each surveying date (k) at each landing site (a):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku} \left(\frac{LF_{aku}}{LF_{aku} + LN_{aku}} \right)}{LF_{aku} - ri_{aku}} \quad (\text{Equation 2})$$

where LF_{aku} was the number of interviewed boats that were fishing (a subset of these, ri_{aku} , refused an interview, or had an incomplete interview that could not be used), LN_{aku} was the number of interviewed boats that were not fishing, and LM_{aku} was the number of boats that were not interviewed by surveyor u , during surveying date k , at landing site a . This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term $A_{mdsakut}$ to denote the number of non-lodge boats reporting activity during time-block t , as interviewed by surveyor u on survey date k , at landing site a , in subarea s , during month m , and on day type d (n_{mdsaku} was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{mdst} = \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot A_{mdsakut}) \quad (\text{Equation 3})$$

Table 1. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns and CPE, for all levels of each factor, 2019.

Month	Day Type	Subarea												TOTALS		
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall
May	WD	7	2	1	45	47	44	11	2	19	2	12	3	10	185	195
	WE	1	2	1	82	57	68	16	0	38	6	10	3	4	280	284
	May Total	8	4	2	127	104	112	27	2	57	8	22	6	14	465	479
June	WD	25	22	6	125	50	69	22	2	100	12	46	8	53	434	487
	WE	27	9	7	100	52	83	28	4	95	12	49	8	43	431	474
	June Total	52	31	13	225	102	152	50	6	195	24	95	16	96	865	961
July	WD	44	33	22	145	54	57	41	2	160	27	56	19	99	561	660
	WE	49	33	11	116	67	95	29	5	133	31	45	16	93	537	630
	July Total	93	66	33	261	121	152	70	7	293	58	101	35	192	1,098	1,290
August	WD	23	17	55	111	48	55	18	1	112	15	27	2	95	389	484
	WE	23	16	29	80	73	59	33	1	103	18	29	13	68	409	477
	August Total	46	33	84	191	121	114	51	2	215	33	56	15	163	798	961
GRAND TOTAL		199	134	132	804	448	530	198	17	760	123	274	72	465	3,226	3,691

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{mds} = \sum_t A'_{mdst} \quad (\text{Equation 4})$$

The proportion of non-lodge boats (P_{mdst}) that were actively fishing during each of the hourly time-blocks (t) was calculated for each month, day type, and subarea:

$$P_{mdst} = \frac{A'_{mdst}}{\sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad (\text{Equation 5})$$

Using this method, 96 unique angler activity patterns were to be estimated (i.e., 4 months $\times$ 2 day types $\times$ 12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 96 blocks. In the end, some blocks contained too few interviews (Table 1), so it was decided to pool activity data over subarea. The equation for non-lodge angler activity was thus:

$$P_{mdt} = \frac{\sum_s A'_{mdst}}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad (\text{Equation 6})$$

and the associated variance was:

$$S_{P_{mdt}}^2 = \frac{(P_{mdt})(1 - P_{mdt})}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad (\text{Equation 7})$$

Catch Per Effort Estimation

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview (i), the guided status (g), month (m), and the catch (C) of each species (r) in each subarea (s) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{msrgi} = \frac{c_{msrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\} \quad (\text{Equation 8})$$

Mean CPE was calculated as the average catch, over all n_{msg} boat trips:

$$C\hat{P}E_{msg} = \frac{\sum_{i=1}^{n_{msg}} c_{msrgi}}{n_{msg}} \quad (\text{Equation 9})$$

The variance for the estimate of mean catch per effort was calculated as:

$$S_{C\hat{P}E_{msg}}^2 = \frac{\sum (CPE_{msrgi} - C\hat{P}E_{msg})^2}{(n_{msg} - 1)} \quad (\text{Equation 10})$$

In the event that a month/subarea-specific sample size was < 5 , adjacent subareas would be pooled together as follows:

- if n for 3I, 3J or 3K was too low, then it was combined with the other two subareas, along with data from ambiguous parts of Area 3 (“3?”) to give a 3?IJK regional average CPE ;
- if n for 4A, 4B or 4C was too low, then it was combined with the other two to give a 4ABC regional average CPE ;
- if n for 4D, 4E or 4F was too low, then it was combined with the other two to give a 4DEF regional average CPE ; and
- if n for 4G, 4H or 4L was too low, then it was combined with the other two to give a 4GHL regional average CPE .

Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip (RPE).

Angler Effort Estimation

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

Lodge Effort Census

A complete census of lodge-based angling effort was collected from lodge log books by DFO personnel (I. Winther, DFO Prince Rupert, pers. comm.). DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodge based vessels were considered unavailable to the flights on change-over days (days when the lodges received new clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight). Change-over days occurred twice a week but timing was often different for each lodge, making 29% (2/7) of the lodge effort not visible to flights due to change-over days. 2) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day. Thus, approximately 60% of the remaining vessel activity was available to the flight counts on ‘regular’ days (i.e., not change-over). 3) Vessels on mid-day break or on change-over days were excluded from the numbers of vessels visible to the flights. All of the

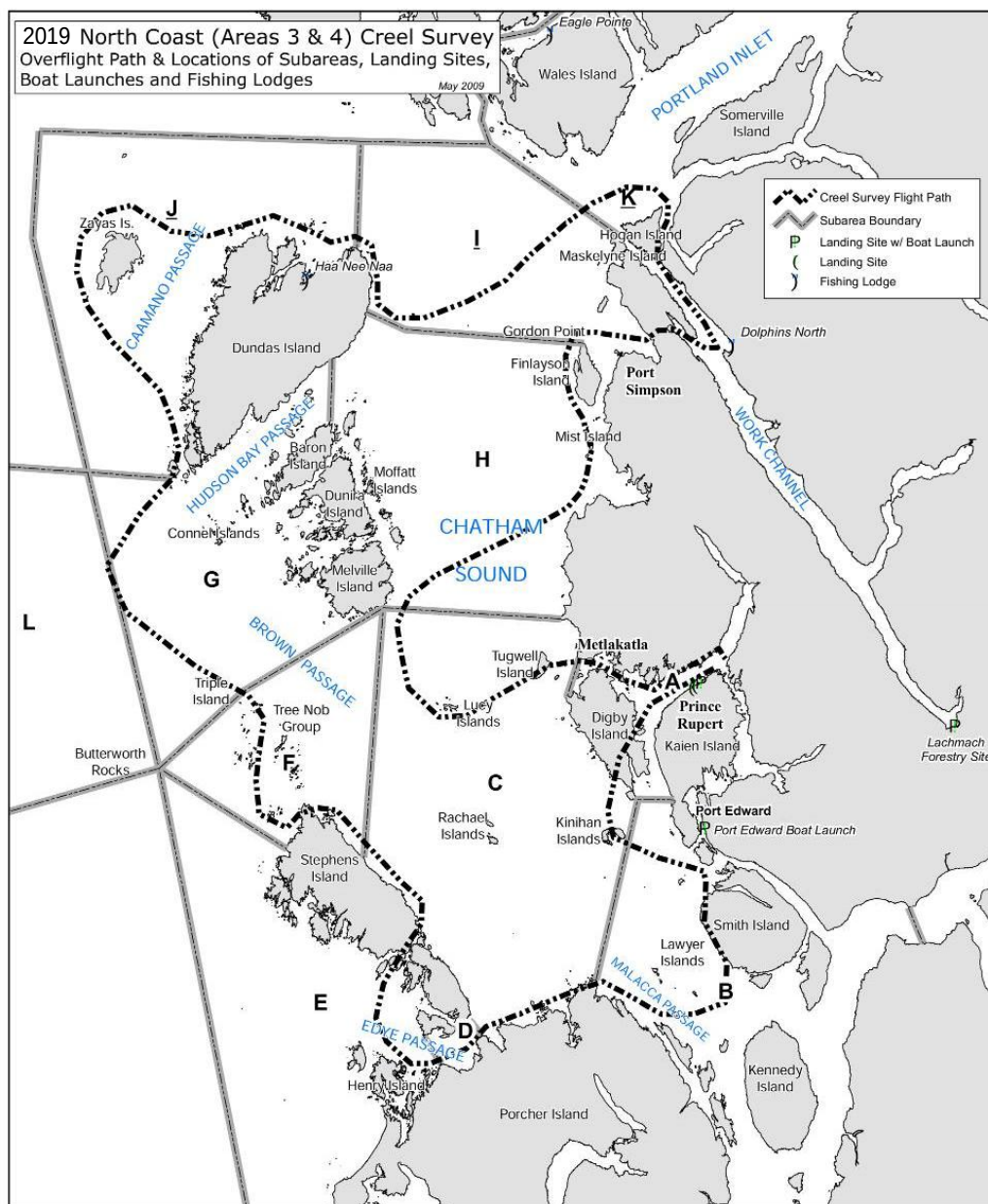


Figure 2. Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

Table 2. Summary of overflight sampling effort (number of flights shifts) by day type and by month, 2019.

Month	Weekday	Weekend/ Holiday	Total
May	5	3	8
June	5	3	8
July	5	3	8
August	5	3	8
Total	20	12	32

lodge catch was reported from Area 3, without subarea-specific effort information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

Non-lodge Effort Surveys

Table 2 shows the number of overflight surveys scheduled for each day type and month. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of small (< 30 ft) and large (≥ 30 ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey o (conducted during month m and on day type d), observers tallied the total number of recreational boats that were actively fishing at time t on their i^{th} pass over subarea s , V_{mdsoit} . For each visit to a subarea within Management Area 3, the relative proportion of boats was calculated, ρ_{mdsoit} , and the total number of lodge boats (λ_{mdo}) apportioned accordingly:

$$\lambda_{mdsoit} = \rho_{mdsoit} \cdot \lambda_{mdo} \quad (\text{Equation 11})$$

The number of non-lodge boats was hence:

$$v_{mdsoit} = V_{mdsoit} - \lambda_{mdsoit} \quad (\text{Equation 12})$$

Since angling occurs over the course of the entire day, the number of non-lodge boats that were observed was divided by the proportion of average daily number of boats active (P_{mdst}) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, thus the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea, B_{mdso} :

$$B_{mdso} = \sum_i \frac{v_{mdsoit}}{P_{mdst}} \quad (\text{Equation 13})$$

The variance of B_{mdso} was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} \text{if } z = x / y, \text{ } Var(z) &= (y^{-2})Var(x) + (x^2 y^{-4})Var(y) ; \\ \text{if } z = xy, \text{ } Var(z) &= (y^2)Var(x) + (x^2)Var(y) ; \text{ and} \\ \text{if } z = cx, \text{ } Var(z) &= (c^2)Var(x) \quad \text{where } c \text{ is a constant.} \end{aligned} \quad (\text{Equation 14})$$

Since there was no variance associated with the boat counts,

$$S_{B_{mdso}}^2 = (v_{mdsoit})^2 (P_{mdst})^{-4} S_{P_{mdst}}^2 \quad (\text{Equation 15})$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys conducted, n_{mds} , as:

$$\hat{B}_{mds} = \frac{\sum_{o=1}^{n_{mds}} B_{mdso}}{n_{mds}} \quad (\text{Equation 16})$$

with a variance of:

$$S_{\hat{B}_{mds}}^2 = (n_{mds})^{-2} S_{B_{mdso}}^2 \quad (\text{Equation 17})$$

Total monthly fishing effort was calculated for each day type and subarea by multiplying the average daily effort by the number days of day type d that occurred in month m :

$$E_{mds} = \hat{B}_{mds} \cdot N_{md} \quad (\text{Equation 18})$$

with a variance of:

$$S_{E_{mds}}^2 = S_{\hat{B}_{mds}}^2 \cdot N_{md}^2 \quad (\text{Equation 19})$$

The standard error of the estimate of the total monthly fishing effort, after pooling over day types, was:

$$S_{E_{ms}} = \sqrt{\frac{\sum_d S_{E_{mds}}^2}{\sum_d n_{mds}}} \quad (\text{Equation 20})$$

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of month and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of month and day type during which the data gap occurred.

Catch and Harvest Estimation

Total catch was calculated for each month, subarea and species by multiplying total angling effort by catch per effort, and then summing over day type:

$$C_{msr} = \sum_d (E_{mds} \cdot C\hat{P}E_{msrg}) \quad (\text{Equation 21})$$

where g = 'all' (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{msr}} = \sqrt{\sum_d \left(E_{mds}^2 \frac{S_{CPE_{msrg}}^2}{n_{msg}} + CPE_{msrg}^2 \frac{S_{E_{mds}}^2}{n_{mds}} + \frac{S_{CPE_{msrg}}^2}{n_{msg}} \frac{S_{E_{mds}}^2}{n_{mds}} \right)} \quad (\text{Equation 22})$$

where n_{mds} and n_{msg} are the sample sizes for E and CPE estimates, respectively.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released by replacing CPE with HPE or with RPE in Equations 21 and 22.

Biosampling

When given permission by the anglers, surveyors collected biological data from Chinook and Coho salmon and Pacific Halibut. All three species were sexed when they were intact. Chinook and Coho salmon were examined for clipped adipose fins (mark rate data used by DFO for coded wire tag analyses) and potentially head tags. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook and Coho salmon (used by DFO to determine age, and stock composition).

Ideally, all anglers would allow their catch to be biosampled, and that the examined fish were an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had surveyors ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol in 2016 so that anglers were asked if their catch could be biosampled before any other interactions took place. Also starting 2016, no fin clip data were accepted on the word

of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

Biomass of Harvested Pacific Halibut

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned}\text{Weight(lb)}_{\text{net}} &= 6.921 \times 10^{-6} \times \text{FL(cm)}^{3.24} \\ \text{Weight(lb)}_{\text{round}} &= 9.166 \times 10^{-6} \times \text{FL(cm)}^{3.24}\end{aligned}\quad (\text{Equation 23})$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

Statistical Analyses

For each fish taxa, differences in CPE among months and between trip-types (guided vs. non-guided) were examined using Kruskal Wallis tests (Zar 1984). The same tests were used to examine the effects of month and day type on effort, and the effects of month on catch and harvest. When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

Trailer Census

Starting at the beginning of April, the total number of parked trailers at the landing locations was recorded daily during the peak of the activity timing window (i.e., between noon and 2 PM; similar to the targeted timing of aerial overflights). Theoretically, trailer counts could be used as an index of fishing effort, are more economical than overflights, and are not weather dependant, though no sub-area-specific effort data are collected. DFO has been collecting trailer count data for years and we are maintaining the time series.

Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. For the regression analysis, the location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively. Regression relationships were determined for each month of study (8 points per month-specific regression) and for the entire (May through August) pooled study period ($n = 32$).

Trailer count data were collected throughout April (i.e., one month prior to the start of the creel surveys) in order to calculate an approximate index of effort, to test whether substantial angling activity is being missed by starting the creel survey in May. For each day of trailer counts, we used the *month-specific* linear regressions (for April trailer counts, we used the May regression coefficients) to calculate what the daily boat counts *would have been* had there been overflights each day. Then, the month-specific activity pattern data (for April trailer counts, we used the May activity pattern) were used to expand the estimated boat counts to calculate total fishery effort by day. Summing over months permitted comparisons of estimated effort in April 2019 to that during the four survey months.

RESULTS

Angler Interviews

Over the four month study period, there were 240 observer shifts, during which 2,592 interviews were conducted, involving 7,699 non-lodge anglers (Table 3; Figure 3). July was the busiest month, with 895 interviews conducted (average 14.9 interviews per shift), which represented 35% of the total number of study period interviews (Table 3).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F being the overwhelming favourite (Table 4). Subareas 4A, 4B, and 4C were the next most likely areas for anglers to have reported activity. The subareas with the lowest number of trips reporting activity were 4E, 4L, 4G, and 3J. Only 95 interviews were of parties that only targeted shellfish, and those were largely in subarea 4A (Table 5).

On average, 27% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 11 of 738 trips (1.5%) interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 28% and 50% of the time, respectively (Table 6).

In all there were 44 refusals (Table 4), or 1.7% refusal rate. The highest refusal rate (2.2%) was at Rushbrooke Boat Launch, where 57% of the refusals occurred. The refusal rates for the Port Edward Boat Launch and the Prince Rupert Yacht Club were 2.0% and 0.5%, respectively.

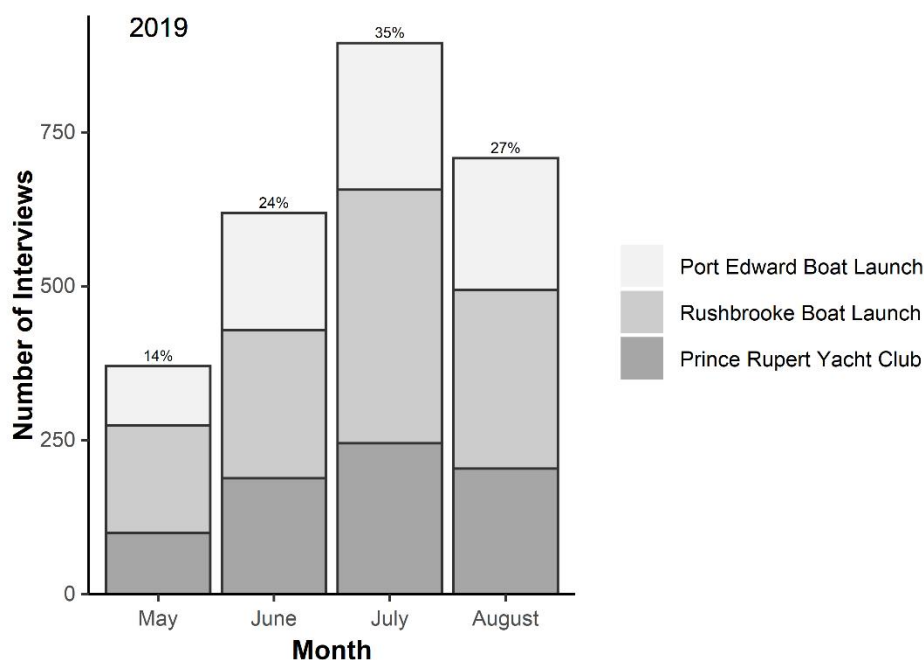


Figure 3. Monthly number of interviews conducted at each of three landing sites, 2019.

Table 3. Monthly number of shifts, and numbers of boats interviewed per landing site, 2019. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.

Landing Site	May				June				July			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	96	226	20	4.8	190	484	20	9.5	238	662	19	12.5
Prince Rupert Yacht Club	99	300	20	5.0	188	606	20	9.4	245	793	21	11.7
Rushbrooke Boat Launch	175	426	20	8.7	241	698	20	12.1	412	1250	20	20.6
TOTAL	370	952	60	6.2	619	1,788	60	10.3	895	2,705	60	14.9
% of Total Interviews		14%				24%				35%		
Estimated Non-lodge Effort		1,391				3,283				4,078		
% of Non-lodge Effort		11%				26%				33%		
% of Effort Sampled		27%				19%				22%		

Landing Site	August				TOTAL			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	214	609	20	10.7	738	1981	79	9.3
Prince Rupert Yacht Club	204	698	20	10.2	736	2397	81	9.1
Rushbrooke Boat Launch	290	947	20	14.5	1118	3321	80	14.0
TOTAL	708	2,254	60	11.8	2,592	7,699	240	10.8
% of Total Interviews		27%						
Estimated Non-lodge Effort		3,644				12,397		
% of Non-lodge Effort		141%						
% of Effort Sampled		19%				21%		

Table 4. Fishing locations as reported by interviewed non-lodge parties, by month and subarea, 2019. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.

Month	Subarea															TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	
May	6	2	2	88	88	85	20	1	44	6	13	4		3	9	370
June	31	19	8	122	77	102	36	4	126	16	53	10		7	10	619
July	65	42	22	146	97	122	54	6	203	35	67	19	1	5	14	895
August	32	25	58	105	102	94	40	1	163	21	38	8	1	8	11	708
TOTAL	135	87	89	461	364	403	149	12	535	78	171	41	2	22	44	2,592

Table 5. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2019.

Month	Subarea															TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	
May	0	0	1	17	4	2	0	0	0	0	0	0	0	0	0	24
June	0	0	0	23	1	0	0	0	0	0	0	0	0	0	0	24
July	0	0	0	26	0	0	0	0	1	0	0	0	0	0	0	27
August	0	0	0	16	1	3	0	0	0	0	1	0	0	0	0	20
TOTAL	0	0	1	82	6	5	0	0	1	0	1	0	0	0	0	95

Table 6. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2019.

Landing Site	Trip Type		% guided
	Guided	Unguided	
Port Edward Boat Launch	11	727	1.5%
Prince Rupert Yacht Club	365	371	50%
Rushbrooke Boat Launch	314	804	28%
TOTAL	690	1,902	27%

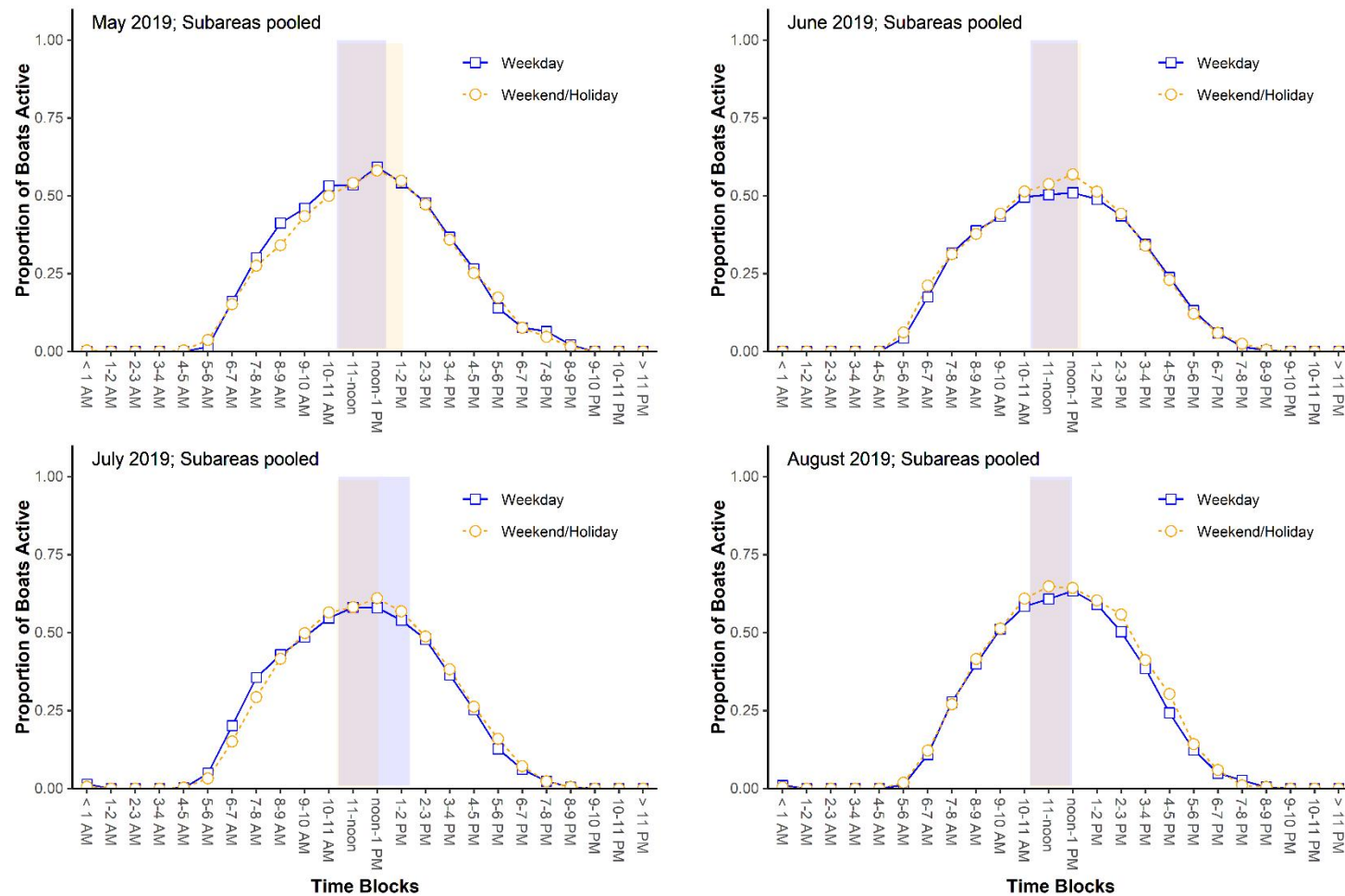


Figure 4. Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and month, pooled over all subareas, 2019. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and month.

Angler Activity Patterns

Non-lodge angler activity pattern estimates varied little among months (Figure 4). There was virtually no difference between the weekday and weekend/holiday activity patterns, except for a slightly greater tendency for weekend anglers to be active during the peak times, as compared to weekday anglers in June and August. Timing of our overflights lined up with peak activity levels in all months.

Catch Per Effort Estimates

Sample sizes were adequate ($n \geq 5$) to produce reliable estimates of non-lodge CPE in most combinations of months and creel study sub-areas, (Table 7) except in subareas 3J, 3K, and 4E in May, and in subarea 4E in August. However, when the interview data were split between guided and non-guided groups, month-specific sample sizes did not suffice in several of the management areas (i.e., 3I-3K, 4B, 4D, 4E, 4G, and 4L; Table 7). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPE calculations (Table 8). Average number of fish harvested per non-lodge boat trip (HPE) was calculated for each species by month, subarea, and by trip type (Appendix 2).

HPE varied significantly among months (Table 9) for Chinook ($\chi^2 = 12.2$, $df = 3$, $P = 0.007$), Chum (*O. keta*, $\chi^2 = 16.3$, $df = 3$, $P = 0.001$), Coho ($\chi^2 = 26.1$, $df = 3$, $P < 0.0001$), and Pink salmon (*O. gorbuscha*, $\chi^2 = 26.9$, $df = 3$, $P < 0.0001$). For Chinook Salmon, median HPE from May and June were significantly lower than in August (Table 9; Figure 5). For Chum Salmon, post-hoc tsts did not reveal any significant among-month pairwise differences (Table 9; Figure 5). For Coho Salmon, median HPE in May was significantly lower than in June, which was significantly lower than in July or August, and the last two months did not differ significantly (Table 9; Figure 5). For Pink Salmon, HPE from May and June were significantly lower than in July or August (Table 9; Figure 5). There were no significant differences among months for HPE of Sockeye salmon (*O. nerka*), or for any species of groundfish (including Pacific Halibut and Lingcod; Figure 5) or rockfish species.

Guided trips had higher HPE than non-guided trips, and the differences were statistically significant for a variety of species (Table 10). In particular, guided trips caught 277% more Chinook Salmon ($\chi^2 = 17.4$, $df = 1$, $P < 0.0001$) and 224% more Pacific Halibut ($\chi^2 = 21.1$, $df = 1$, $P < 0.0001$) than unguided trips. Although guided trips caught 213% more Coho Salmon and 161% more Lingcod than unguided trips, the differences were not statistically significant (Coho Salmon: $\chi^2 = 1.5$, $df = 1$, $P = 0.23$; Lingcod: $\chi^2 = 2.1$, $df = 1$, $P = 0.14$). Greenling and Rock Sole also had significantly higher HPE in guided trips, but the differences were not reflected in the median values (Table 10).

For lodge boat trips, HPE (and RPE for Chinook Salmon) estimates are shown in Table 11.

Table 7. Sample sizes for CPE (harvest per effort) estimates by month, for guided and not guided non-lodge boat trips, 2019.

Period	Trip Type	Subarea											
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
May	Guided	2	3	0	30	7	19	5	1	16	3	8	5
	Unguided	6	1	2	97	97	93	22	1	41	5	14	1
	Combined	8	4	2	127	104	112	27	2	57	8	22	6
June	Guided	28	18	6	110	4	12	10	0	63	11	40	12
	Unguided	24	13	7	115	98	140	40	6	132	13	55	4
	Combined	52	31	13	225	102	152	50	6	195	24	95	16
July	Guided	45	47	20	134	5	13	9	1	100	22	44	28
	Unguided	48	19	13	127	116	139	61	6	193	36	57	7
	Combined	93	66	33	261	121	152	70	7	293	58	101	35
August	Guided	17	20	50	91	10	13	3	0	71	6	17	14
	Unguided	29	13	34	100	111	101	48	2	144	27	39	1
	Combined	46	33	84	191	121	114	51	2	215	33	56	15

Table 8. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each month and subarea, for guided and not guided non-lodge boat trips, 2019.

Month	Trip Type	Subarea											
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
May	Guided	3?,I,J,K	3?,I,J,K	3?,I,J,K	4A	4B	4C	4D	4D,E,F	4F	4G,H,L	4H	4L
	Unguided	3I	3?,I,J,K	3?,I,J,K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3I	3?,I,J,K	3?,I,J,K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
June	Guided	3I	3J	3K	4A	4A,B,C	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
July	Guided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
August	Guided	3I	3J	3K	4A	4B	4C	4D,E,F	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L

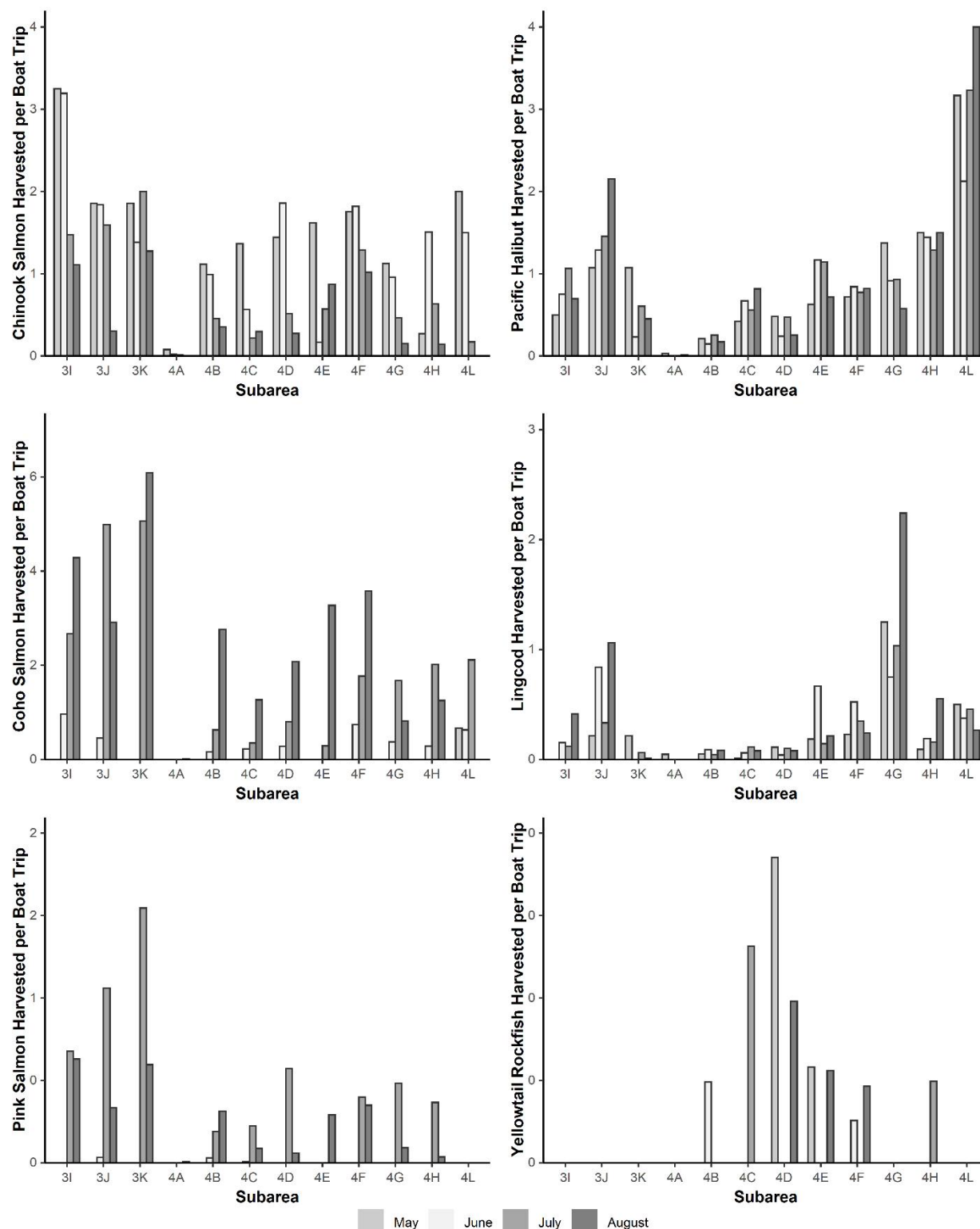


Figure 5. Average number of fish harvested per non-lodge boat trip, for six sportfish species, by month and subarea, 2019. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

Table 9. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among months, by species, 2019. Subareas treated as replicates.

Species	Kruskal-Wallis Statistics			Median HPE			
	χ^2	df	P	May	June	July	August
Chinook Salmon	12.2	3	0.007	1.5	1.4	0.5	0.3
Chum Salmon	16.3	3	0.001	0	0	0	0
Coho Salmon	26.1	3	< 0.0001	0	0.3	1.7	2.4
Pink Salmon	26.9	3	< 0.0001	0	0	0.4	0.2
Sockeye Salmon	3.0	3	0.39	0	0	0	0
Arrow Flounder	0.8	3	0.85	0	0	0	0
Big Skate	3.0	3	0.39	0	0	0	0
Cabezon	0.6	3	0.90	0	0	0	0
English Sole	3.8	3	0.28	0	0	0	0
Greenling	5.5	3	0.14	0	0	0	0.004
Lingcod	0.6	3	0.90	0.1	0.2	0.1	0.2
Pacific Cod	3.4	3	0.33	0	0.02	0.03	0.03
Pacific Halibut	0.2	3	0.98	0.7	0.8	0.9	0.7
Ratfish	0.0	3	1	0	0	0	0
Rock Sole	2.0	3	0.58	0	0	0	0
Sablefish	3.0	3	0.39	0	0	0	0
Starry Flounder	6.3	3	0.10	0	0	0	0
Black Rockfish	3.8	3	0.29	0.02	0	0.05	0.06
Boccacio Rockfish	0.0	3	1	0	0	0	0
Canary Rockfish	1.0	3	0.79	0	0	0	0
China Rockfish	7.2	3	0.07	0	0	0.007	0.004
Copper Rockfish	7.2	3	0.07	0	0.005	0.002	0.008
Dusky Rockfish	0.0	3	1	0	0	0	0
Quillback Rockfish	0.5	3	0.93	0.07	0.07	0.06	0.08
Redband Rockfish	3.0	3	0.39	0	0	0	0
Redstripe Rockfish	3.0	3	0.39	0	0	0	0
Tiger Rockfish	3.4	3	0.34	0	0	0	0
Widow Rockfish	0.0	3	1	0	0	0	0
Yelloweye Rockfish	0.0	3	1	0	0	0	0
Yellowtail Rockfish	0.4	3	0.94	0	0	0	0

Table 10. Kruskal-Wallis tests for differences in HPE (fish caught per non-lodge boat trip) between guided and non-guided boats, by species, 2019. Months and subareas treated as replicates.

Species	Kruskal-Wallis Statistics			Median HPE	
	χ^2	df	P	Guided	Not
Chinook Salmon	17.4	1	< 0.0001	1.8	0.6
Chum Salmon	1.0	1	0.32	0	0
Coho Salmon	1.5	1	0.23	0.7	0.3
Pink Salmon	0.8	1	0.37	0	0
Sockeye Salmon	1.0	1	0.32	0	0
Arrow Flounder	1.8	1	0.18	0	0
Big Skate	1.0	1	0.32	0	0
Cabezon	0.7	1	0.41	0	0
English Sole	0.3	1	0.58	0	0
Greenling	11.3	1	0.0008	0	0
Lingcod	2.1	1	0.14	0.2	0.1
Pacific Cod	0.3	1	0.56	0	0
Pacific Halibut	21.1	1	< 0.0001	1.4	0.6
Ratfish	0.0	1	1	0	0
Rock Sole	4.1	1	0.042	0	0
Sablefish	1.0	1	0.32	0	0
Starry Flounder	1.0	1	0.33	0	0
Black Rockfish	0.5	1	0.46	0	0.02
Boccacio Rockfish	0.0	1	1	0	0
Canary Rockfish	0.3	1	0.57	0	0
China Rockfish	2.7	1	0.10	0	0
Copper Rockfish	0.0	1	0.87	0	0
Dusky Rockfish	0.0	1	1	0	0
Quillback Rockfish	1.3	1	0.25	0.10	0.04
Redband Rockfish	1.0	1	0.32	0	0
Redstripe Rockfish	1.0	1	0.32	0	0
Tiger Rockfish	1.4	1	0.23	0	0
Widow Rockfish	0.0	1	1	0	0
Yelloweye Rockfish	0.0	1	1	0	0
Yellowtail Rockfish	0.9	1	0.35	0	0

Table 11. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, by month, from the census of Area 3 lodges, 2019.

Species	May	June	July	August	Total
Chinook Salmon:					
retained	1.6	1.7	1.4	0.3	1.2
released	0.6	0.8	0.5	0.2	0.5
Coho Salmon	0	0.3	2.7	2.9	1.8
Pink Salmon	0	0.001	0.08	0.05	0.04
Chum Salmon	0	0	0	0.02	0.005
Sockeye Salmon	0	0	0	0	0
Pacific Halibut	0.4	0.3	0.7	0.5	0.5
Lingcod	0.3	0.3	0.6	0.3	0.4
Other groundfish	0	0	0	0	0
Yelloweye Rockfish	0	0	0	0	0
Other rockfish	0.3	0.3	0.5	0.5	0.4

Angler Effort Estimates

Over the four month study period, a total of 32 effort surveys were conducted, including 20 weekday surveys and 12 conducted on weekends/holidays (Table 2, Appendix 3). The numbers of lodge boats that were visible during the overflights (Appendix 3) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts (“the non-lodge small boat counts”) were significantly correlated to trailer counts at both survey launch sites (Rushbrooke: $F_{1,29} = 92.8$, $P < 0.0001$; Port Edwards: $F_{1,29} = 94.6$, $P < 0.0001$), and when the two launch sites were combined ($F_{1,29} = 134.1$, $P < 0.0001$). The fit of the individual-sites regressions (Rushbrooke $R^2 = 0.75$; Port Edward $R^2 = 0.76$) were lower than that of the combined-site regression ($R^2 = 0.82$; Figure 6).

The non-lodge boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2019 (Table 12, Figures 7 and 8). Median daily non-lodge effort on weekdays (4.9 trips per day per subarea) was lower than on weekends/holidays (7.2 trips per day per subarea), but the effect was not statistically significant

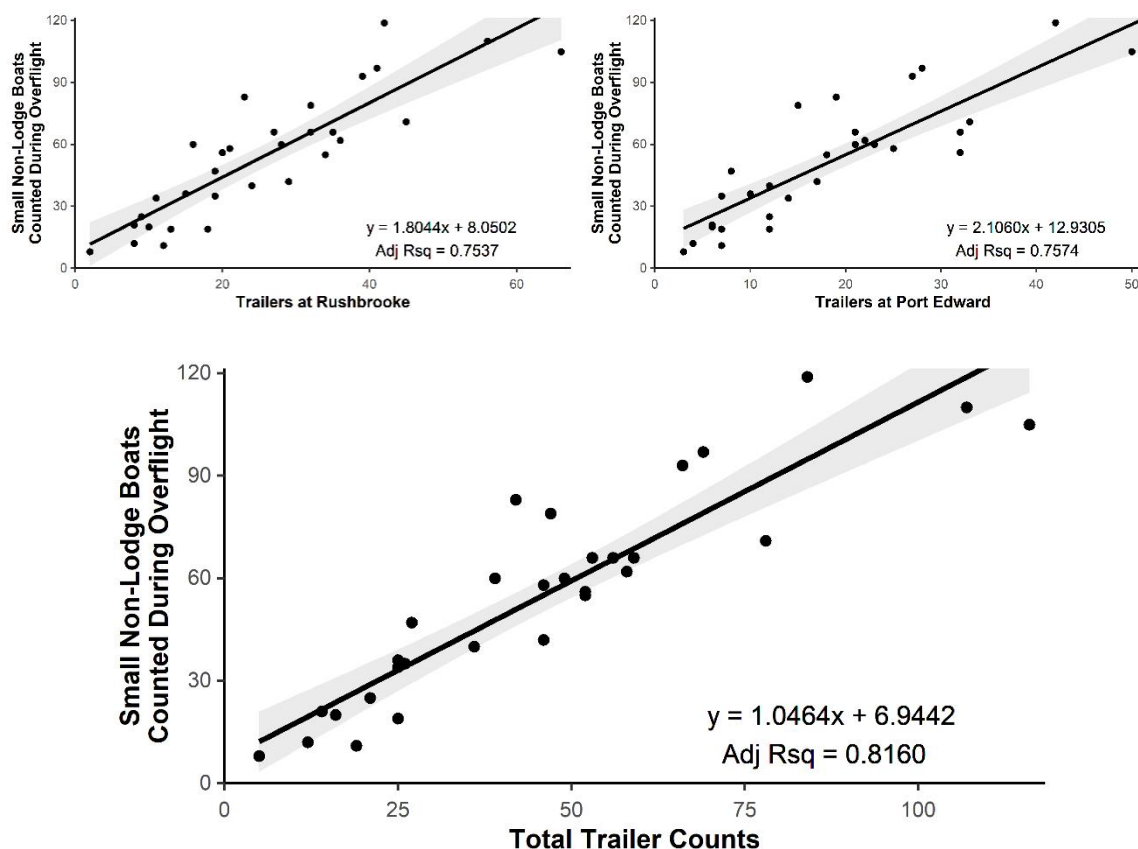


Figure 6. Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded area shows the 95% confidence interval of the predicted values. Note that X axis scale varies among plots.

Table 12. Lodge, non-lodge and total effort estimates (boat trips per month), by month and subarea, 2019.

Month		Subarea												TOTALS			SE of the Totals		
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May	Creel Estimate	84.4	89.2	103.1	133.4	204.3	433.8	38.2	41.5	142.8	35.9	40.6	44.1	277	1,115	1,391	3	9	10
	Lodge Report	49.4	52.2	60.4										162	0	162			
	TOTAL MAY	133.8	141.4	163.5	133.4	204.3	433.8	38.2	41.5	142.8	35.9	40.6	44.1	439	1,115	1,553	3	9	10
	SE	1.7	2.1	2.1	2.3	4.3	6.2	1.1	1.6	3.8	1.0	0.9	1.6						
June	Creel Estimate	324.6	431.6	343.1	79.1	360.3	566.3	101.7	49.6	583.8	172.4	125.2	145.6	1,099	2,184	3,283	9	12	15
	Lodge Report	149.7	199.1	158.2										507	0	507			
	TOTAL JUNE	474.3	630.6	501.3	79.1	360.3	566.3	101.7	49.6	583.8	172.4	125.2	145.6	1,606	2,184	3,790	9	12	15
	SE	4.6	6.1	4.5	0.9	5.0	5.1	2.0	1.4	7.9	3.5	1.9	2.6						
July	Creel Estimate	428.6	672.0	268.0	99.1	356.9	750.1	92.3	71.3	878.5	262.8	141.9	56.9	1,369	2,710	4,078	9	11	14
	Lodge Report	177.2	277.9	110.9										566	0	566			
	TOTAL JULY	605.9	949.9	378.9	99.1	356.9	750.1	92.3	71.3	878.5	262.8	141.9	56.9	1,935	2,710	4,644	9	11	14
	SE	3.9	7.6	2.6	1.0	3.7	4.6	1.4	0.8	8.2	3.1	1.8	1.0						
August	Creel Estimate	650.8	419.9	311.9	114.6	283.7	690.4	65.7	128.3	522.4	204.1	235.0	17.3	1,383	2,262	3,644	9	10	13
	Lodge Report	249.9	161.3	119.8										531	0	531			
	TOTAL AUGUST	900.7	581.2	431.7	114.6	283.7	690.4	65.7	128.3	522.4	204.1	235.0	17.3	1,914	2,262	4,175	9	10	13
	SE	7.0	4.6	3.3	1.0	3.3	5.2	1.0	1.5	6.1	2.5	2.6	0.3						
FULL SEASON TOTAL		2,115	2,303	1,475	426	1,205	2,441	298	291	2,127	675	543	264	5,893	8,270	14,163	16	21	26
FULL SEASON SE		9	11	7	3	8	11	3	3	13	5	4	3						

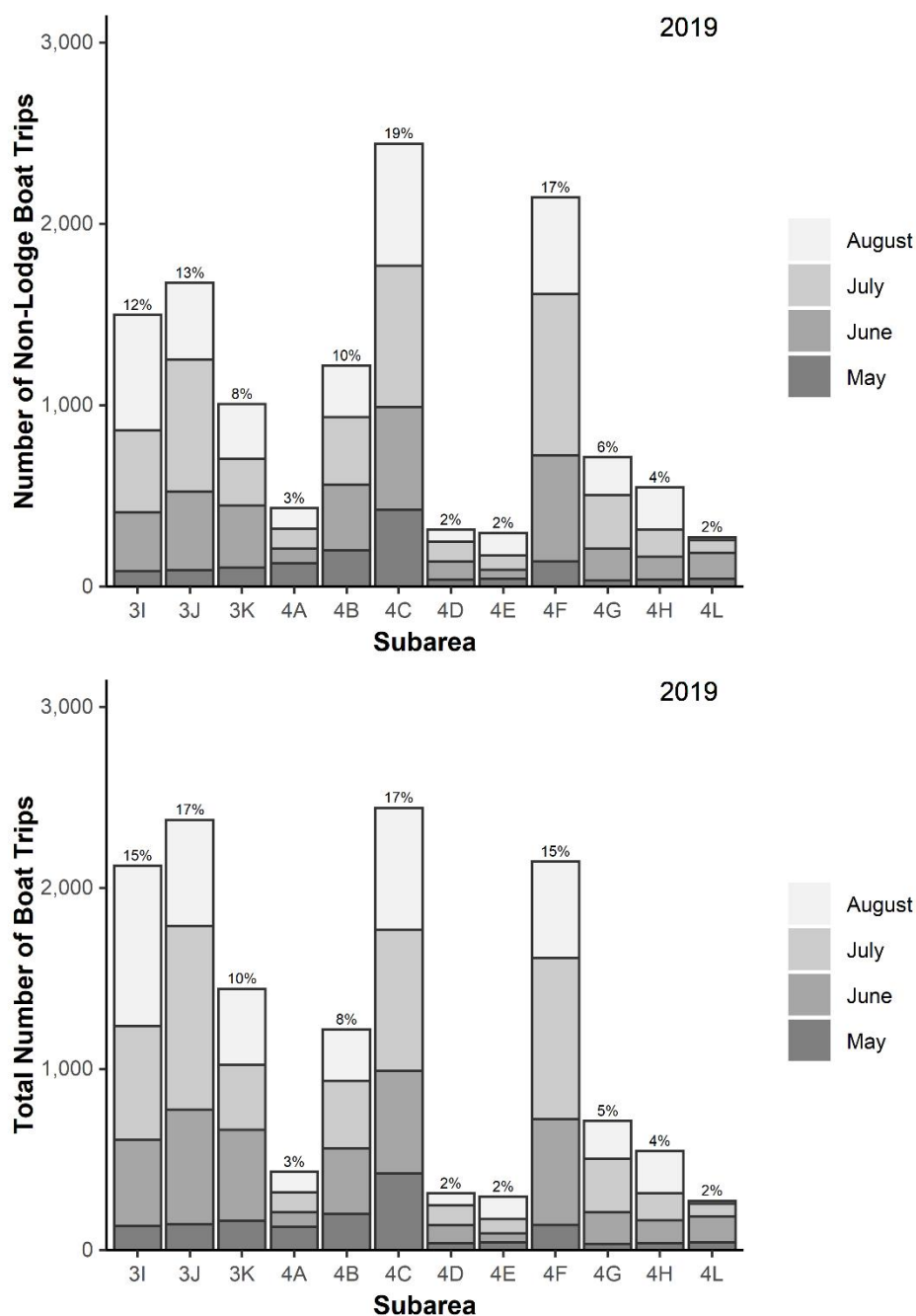


Figure 7. Total estimated numbers of boat trips, by month and subarea, 2019. Upper panel: estimated monthly non-lodge boat trips. Lower panel: total effort, calculated as the sum of lodge and non-lodge boat trips.

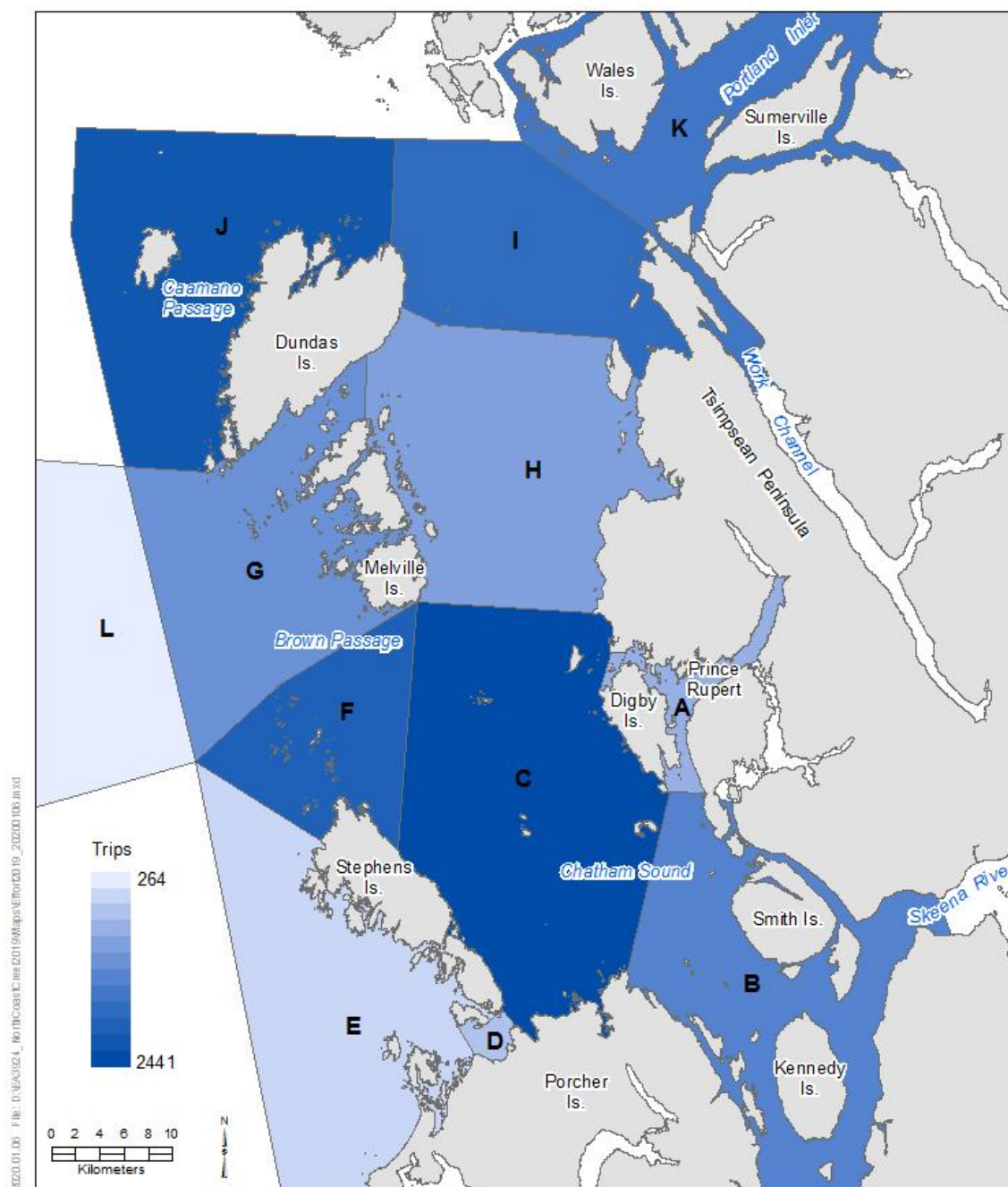


Figure 8. Heat map showing the estimated total effort (numbers of boat trips, lodge + non-lodge) in each subarea in 2019. See Table 12 for details.

($\chi^2 = 2.2$, $df = 1$, $P = 0.14$). Effort varied significantly among months, ranging from 2.9 trips per day per subarea in May to 8.0 to 8.5 trips per day per subarea in June, July and August ($\chi^2 = 16.3$, $df = 3$, $P = 0.001$).

Total lodge effort in 2019 was 162, 507, 566, and 531 boat trips in May, June, July and August, respectively (Table 12). To calculate total (lodge + non-lodge) fishing effort by subarea, we

distributed the lodge boat trips among the three subareas in Management Area 3, according to the relative non-lodge effort in each (Table 12; Figure 7). In all, there were 1,553 (SE = 10) boat trips in May, 3,790 (SE = 15) boat trips in June, 4,644 (SE = 14) boat trips in July, and 4,175 (SE = 13) boat trips in August. The full-season (May-August) Area 3/4 effort estimate for 2019 was 14,163 boat trips (SE = 26). For comparison with our previous study years, which did not include surveys in May, the June-August effort total was 12,610 boat trips (SE = 24).

April Effort

Trailer counts began in April of 2019 and ran until the end of the creel monitoring period (Figure 9). Month-specific boat-count-to-trailer-count regression lines were used to calculate what the daily boat counts *would have been* had there been overflights each day. And the month-specific and day-type-specific activity pattern data (Figure 4) were used to expand the boat counts to calculate total fishery effort by day (Figure 9). Summing over months revealed that total effort in April 2019 was equivalent to 65% of that estimated for May (about 1,000 trips).

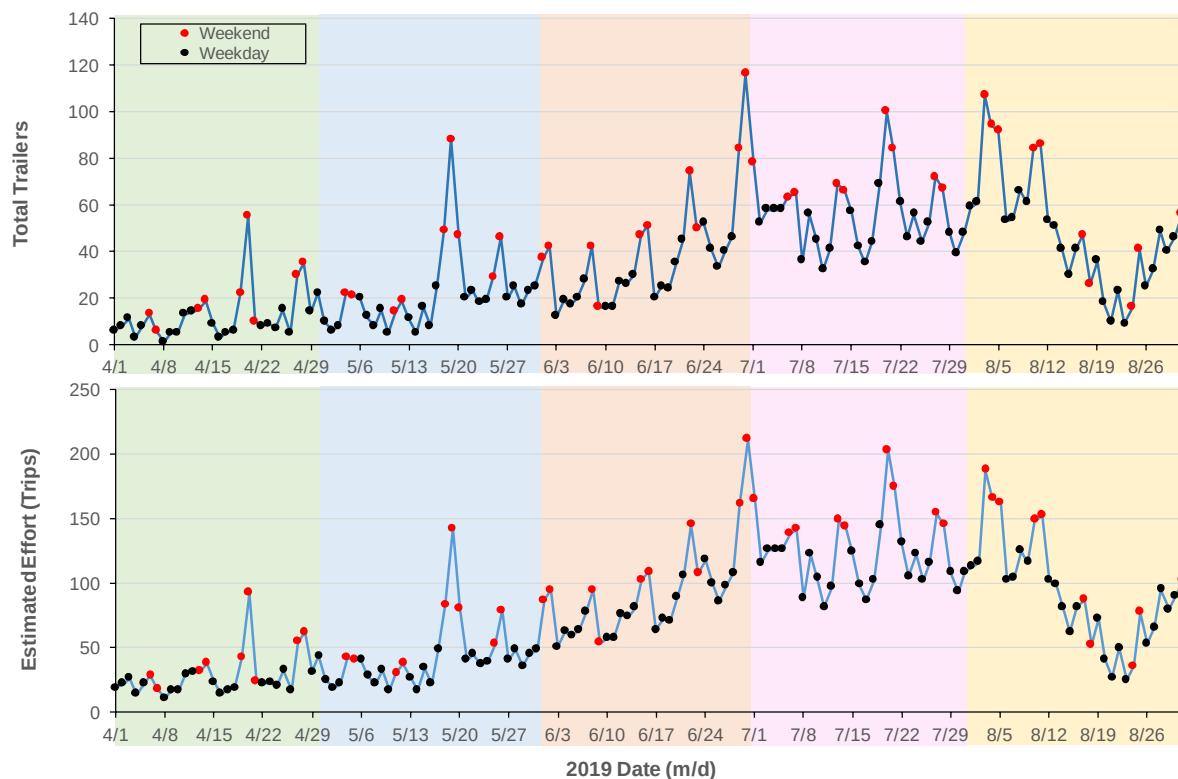


Figure 9. Total number of trailers counted at landing sites (upper), and total fishery effort (lower), by date, April through August. Background colours denote months. Lower: estimates made use of boat-count-from-trailer-count regression and angler activity profiles.

Catch Estimates

Monthly and full-season estimates were generated of the numbers of fish released and retained, by species (Tables 13-22; Appendix 4). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 15,152 (SE = 330) Chinook Salmon were retained by anglers in Areas 3 or 4 during the 2019 study period (Table 14; Figure 10). The largest proportional harvest came from subarea 3I (23%), followed by 4F (20%), 3J (19%) and 3K (19%). Harvest differed markedly over the study period, though the among-month differences were not statistically significantly lower ($\chi^2 = 3.5$, $df = 3$, $P = 0.32$; Table 23). During the study, the number of released Chinook Salmon was estimated at 11,129 (SE = 461), or 42% of the total catch. During May and August, retention (calculated for non-lodge anglers) was 15-16%, in contrast to 32-37% during June and July.

The salmonid that was harvested in the greatest numbers during the 2019 study period was, by far, Coho Salmon (23,820 fish retained, SE = 550; Table 15; Figure 11). The largest proportional harvest came from subareas 3J (25%) and 3I (22%), followed by subareas 3K and 4F (16-17%). Harvest differed significantly among months: harvest in May was significantly lower than that in June, which was in turn significantly lower than in July or August ($\chi^2 = 24.8$, $df = 3$, $P < 0.0001$; Table 23). During the study period, an estimated 2,107 (SE = 179) Coho Salmon were released by anglers (8% of the total catch).

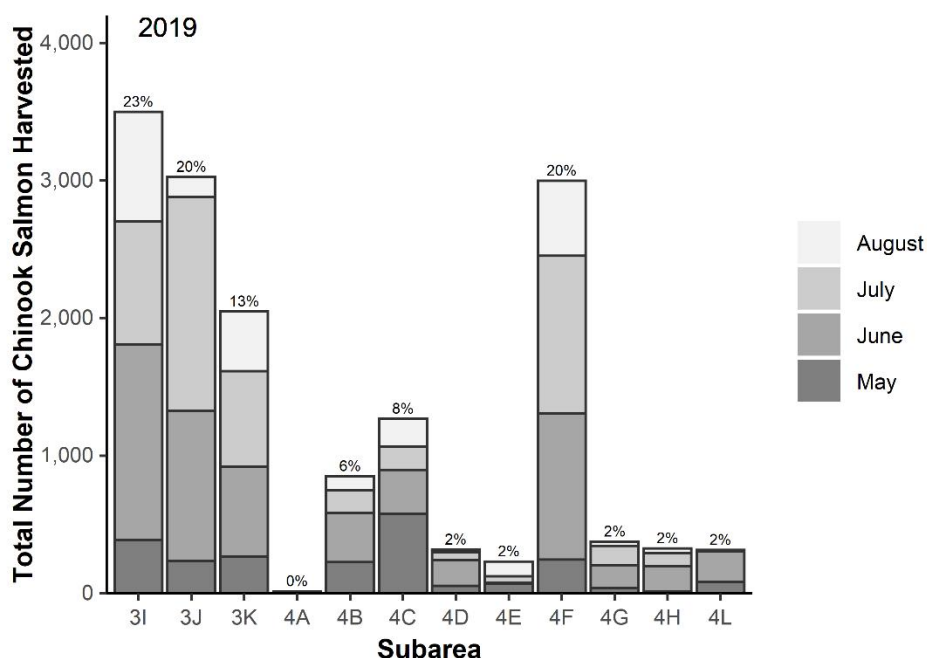


Figure 10. Total (lodge + non-lodge) estimated Chinook Salmon harvest, by month and subarea, 2019.

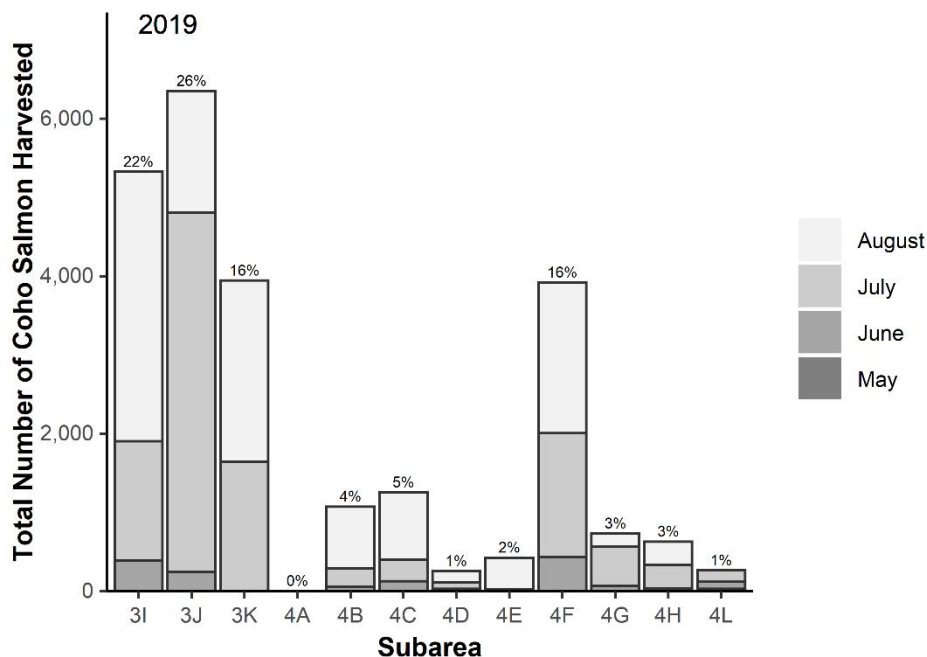


Figure 11. Total (lodge + non-lodge) estimated Coho Salmon harvest, by month and subarea, 2019.

Pink salmon was the least likely salmonid to be retained in 2019 (Table 16). We estimate that 8,956 (SE = 504) fish were caught, of which 39% were retained (3,471 fish, SE = 253), and 610% were released (5,485 fish, SE = 435). Harvest of Pink Salmon varied significantly among months ($\chi^2 = 26.1$, $df = 3$, $P < 0.0001$; Table 23), with significantly lower harvest in May or June than in July or August. Very few Chum (62 fish, SE = 20) or Sockeye (28 fish, SE = 20) salmon were harvested in Area 3 and 4 during the study period (Tables 17 and 18).

An estimated 10,967 (SE = 290) Pacific Halibut were harvested in Areas 3 or 4 during the 2019 study period (Table 19; Figure 12). Most of the harvest could be attributed to effort in subareas 3I, 3J, 4C, and 4F (together 70%). Harvest varied among months (lowest in May, highest in July), but the effect was not statistically significant ($\chi^2 = 6.9$, $df = 3$, $P = 0.07$; Table 23). We estimated that 7,484 (SE = 661) Pacific Halibut were released (41% of the total catch).

From May to August 2019, an estimated 4,514 (SE = 252) Lingcod were harvested, with lowest median harvest in May, and highest in August, though the differences were not statistically significant. ($\chi^2 = 6.0$, $df = 3$, $P = 0.11$; Table 20; Figure 13). The harvest occurred largely in subarea 3J (35%), 4G (20%), and 4F (17%). Of the captured Lingcod, 14% were released (752 fish, SE = 88).

An estimated 3,182 (SE = 267) other ground fish were caught from May to August 2019, of which 68% was released after capture. During the study period, an estimated 1,007 (SE = 117) individuals were retained (Table 21), mostly (58%) Pacific cod (*Gadus macrocephalus*), but also greenlings (Hexagrammidae, 16%), and Arrow Flounder (*Atheresthes stomias*, 8%).

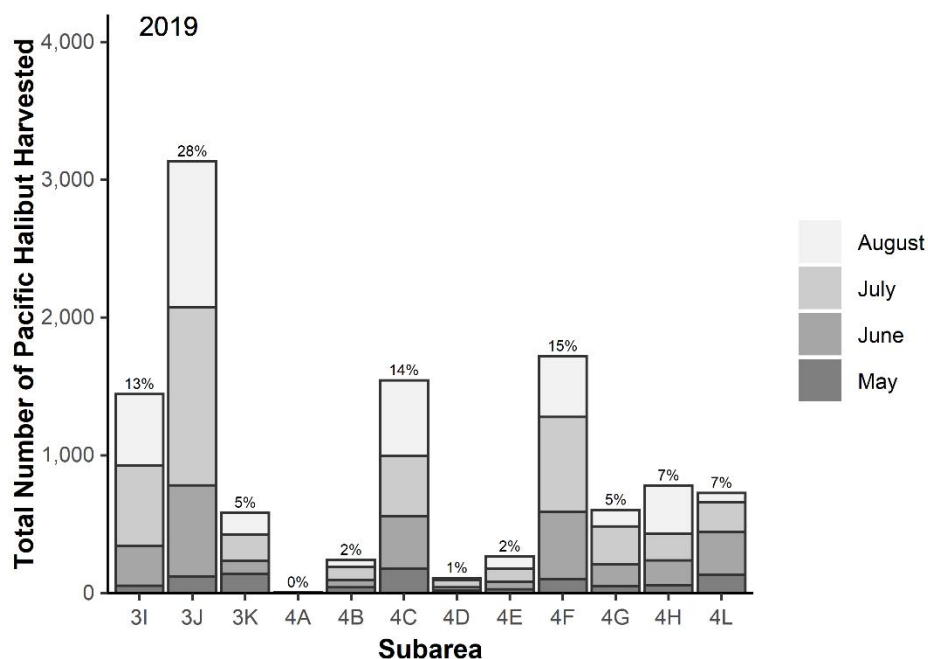


Figure 12. Total (lodge + non-lodge) estimated Pacific Halibut harvest, by month and subarea, 2019.

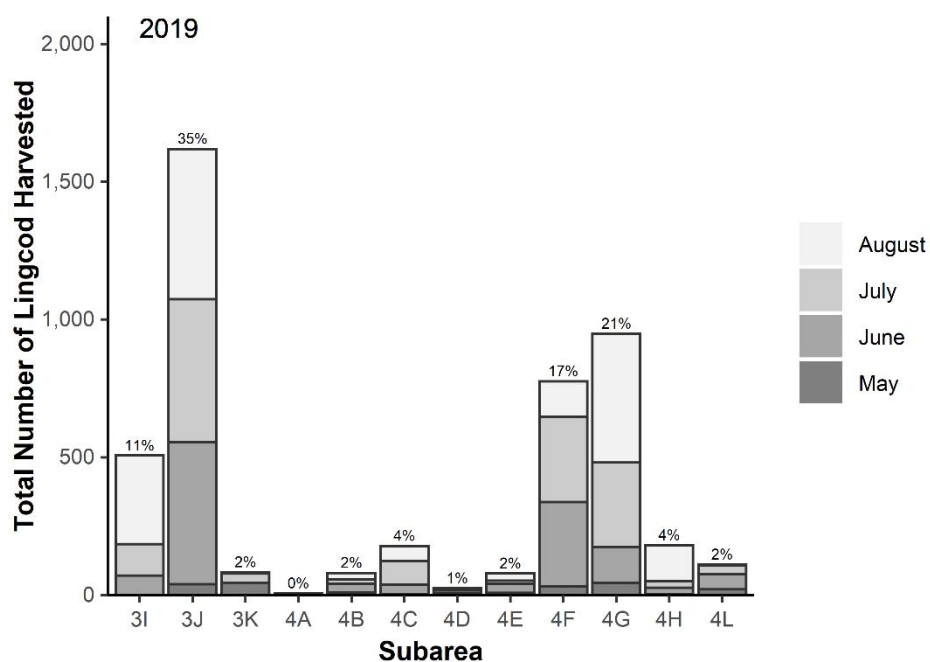


Figure 13. Total (lodge + non-lodge) estimated Lingcod harvest, by month and subarea, 2019.

Table 13. Total (lodge + non-lodge) estimated harvest, by subarea, for popular sportfish taxa, 2019.

Species	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
Chinook Salmon	3,479	2,917	2,098	14	848	1,282	310	228	2,979	359	323	316	8,494	6,658	15,152	271	188	330
Coho Salmon	5,339	5,998	4,095	1	1,064	1,260	239	440	3,856	671	616	241	15,432	8,387	23,820	495	241	550
Pink Salmon	724	895	620	1	168	232	57	37	533	145	60	0	2,238	1,233	3,471	240	81	253
Chum Salmon	17	0	32	0	0	0	0	1	7	0	4	0	49	13	62	19	6	20
Sockeye Salmon	28	0	0	0	0	0	0	0	0	0	0	0	28	0	28	20	0	20
Unknown Salmon	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Pacific Halibut	1,431	3,037	598	6	237	1,545	103	257	1,704	570	777	702	5,066	5,901	10,967	238	167	290
Lingcod	511	1,593	86	6	80	176	23	78	770	904	180	107	2,190	2,324	4,514	203	150	252
other groundfish *	55	236	0	4	84	259	20	16	150	78	88	17	291	717	1,007	84	81	117
Rockfish †	647	1,458	113	5	174	618	179	234	1,033	917	313	96	2,218	3,569	5,787	173	193	260

* includes Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and unknown groundfish

† includes Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

Table 14. Total (lodge + non-lodge) estimated Chinook Salmon catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	386.8	233.6	270.1	10.5	227.9	592.6	55.2	67.0	250.4	40.4	11.1	88.3	891	1,343	2,234	85	80	117
Released	272.5	164.6	190.3	10.5	94.3	220.8	9.9	23.2	102.7	18.0	1.8	36.8	627	518	1,145	67	47	81
June																		
Retained	1422.4	1089.3	652.1	1.8	356.8	320.4	189.1	8.3	1062.8	165.2	188.5	218.4	3,164	2,511	5,675	165	129	209
Released	1152.8	711.5	1042.2	1.1	219.0	346.5	85.4	0	625.7	114.9	235.9	200.2	2,906	1,829	4,735	281	131	310
July																		
Retained	856.9	1451.0	727.6	1.5	162.2	162.8	47.4	40.8	1133.4	122.3	89.9	9.8	3,036	1,770	4,806	149	89	174
Released	326.1	463.1	687.6	1.1	106.2	202.3	60.6	10.2	437.8	45.3	33.7	8.1	1,477	905	2,382	109	75	133
August																		
Retained	813.0	143.4	447.7	0	100.8	205.9	18.0	111.6	532.1	30.9	33.6	0	1,404	1,033	2,437	130	67	146
Released	802.3	654.9	530.4	0	117.2	169.6	12.9	78.1	371.7	61.9	67.1	0	1,988	878	2,866	291	84	303
TOTAL																		
Retained	3,479	2,917	2,098	14	848	1,282	310	228	2,979	359	323	316	8,494	6,658	15,152	271	188	330
Released	2,554	1,994	2,450	13	537	939	169	111	1,538	240	339	245	6,998	4,130	11,129	425	179	461
Catch	6,033	4,911	4,548	26	1,384	2,221	479	339	4,517	599	662	561	15,492	10,788	26,281	504	260	567

Table 15. Total (lodge + non-lodge) estimated Coho Salmon catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	29.4	0	29	29	0	21	21
Released	0	0	0	0	0	0	0	0	0	0	0	44.1	0	44	44	0	32	32
June																		
Retained	392.8	245.3	0	0	56.5	126.7	28.5	0	434.1	64.6	35.6	91.0	638	837	1,475	89	77	117
Released	31.2	111.4	0	0	7.1	33.5	14.2	0	56.9	14.4	2.6	0	143	129	271	55	23	59
July																		
Retained	1438.4	4215.7	1707.1	0	224.2	261.5	73.8	20.4	1553.2	439.5	286.6	120.3	7,361	2,979	10,341	359	137	384
Released	161.3	203.6	138.1	0	38.3	19.7	9.2	0	164.9	0	59.0	14.6	503	306	809	111	43	119
August																		
Retained	3507.5	1537.5	2388.1	0.6	783.0	872.1	136.6	420.0	1868.4	167.0	293.8	0	7,433	4,541	11,974	329	182	375
Released	367.8	63.6	63.1	0	103.2	157.5	5.2	34.0	162.8	0	25.2	0	495	488	982	102	57	117
TOTAL																		
Retained	5,339	5,998	4,095	1	1,064	1,260	239	440	3,856	671	616	241	15,432	8,387	23,820	495	241	550
Released	560	379	201	0	149	211	29	34	385	14	87	59	1,140	966	2,107	160	81	179
Catch	5,899	6,377	4,296	1	1,212	1,471	268	474	4,240	686	703	299	16,573	9,354	25,926	520	254	579

Table 16. Total (lodge + non-lodge) estimated Pink Salmon catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	0	14.4	0	0	10.6	3.7	0	0	0	0	0	0	14	14	29	10	6	12
Released	0	0	0	0	0	0	0	0	0	14.4	0	0	0	14	14	0	10	10
July																		
Retained	299.9	736.1	427.8	0	67.8	167.8	52.7	0	350.8	126.9	52.0	0	1,464	818	2,282	160	70	175
Released	705.1	1293.1	991.0	0	85.5	177.6	46.1	0	368.8	108.7	54.8	0	2,989	842	3,831	377	101	390
August																		
Retained	423.7	144.6	191.8	0.6	89.1	60.6	3.9	37.4	182.2	18.6	8.4	0	760	401	1,161	178	41	183
Released	396.1	407.2	382.5	0	164.1	30.3	24.5	43.6	174.9	12.4	4.2	0	1,186	454	1,640	186	54	194
TOTAL																		
Retained	724	895	620	1	168	232	57	37	533	145	60	0	2,238	1,233	3,471	240	81	253
Released	1,101	1,700	1,373	0	250	208	71	44	544	135	59	0	4,175	1,310	5,485	420	115	435
Catch	1,825	2,595	1,993	1	417	440	127	81	1,077	281	119	0	6,413	2,543	8,956	484	141	504

Table 17. Total (lodge + non-lodge) estimated Chum Salmon catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
August																		
Retained	17.3	0	31.8	0	0	0	0	1.4	7.3	0	4.2	0	49	13	62	19	6	20
Released	0	0	3.7	0	0	0	0	0	0	0	8.4	0	4	8	12	3	5	5
TOTAL																		
Retained	17	0	32	0	0	0	0	1	7	0	4	0	49	13	62	19	6	20
Released	0	0	4	0	0	0	0	0	0	0	8	0	4	8	12	3	5	5
Catch	17	0	36	0	0	0	0	1	7	0	13	0	53	21	74	19	7	21

Table 18. Total (lodge + non-lodge) estimated Sockeye Salmon catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	1.3	0	0	0	0	0	0	1	1	0	1	1
August																		
Retained	28.3	0	0	0	0	0	0	0	0	0	0	0	28	0	28	20	0	20
Released	0	0	0	0	2.3	6.1	0	1.0	4.9	0	0	0	0	14	14	0	6	6
TOTAL																		
Retained	28	0	0	0	0	0	0	0	0	0	0	0	28	0	28	20	0	20
Released	0	0	0	0	2	6	1	1	5	0	0	0	0	16	16	0	6	6
Catch	28	0	0	0	2	6	1	1	5	0	0	0	28	16	44	20	6	21

Table 19. Total (lodge + non-lodge) estimated Pacific Halibut catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	53.4	121.0	139.9	4.2	43.2	182.1	18.4	26.0	102.7	49.4	60.9	139.7	314	627	941	29	42	51
Released	10.6	38.2	44.2	18.9	3.9	38.7	14.2	12.1	37.6	35.9	12.9	58.8	93	233	326	27	52	58
June																		
Retained	288.9	660.7	93.9	0	53.0	380.0	24.4	57.9	491.0	158.0	180.6	309.4	1,044	1,654	2,698	109	93	143
Released	93.6	988.5	26.4	0	113.0	149.0	2.0	33.1	425.1	64.6	106.8	327.6	1,109	1,221	2,330	403	178	441
July																		
Retained	562.0	1204.0	200.1	0	91.4	419.4	43.5	81.5	680.6	244.7	182.7	183.7	1,966	1,928	3,894	139	100	171
Released	502.3	946.9	56.9	0	88.5	143.1	7.9	214.0	494.7	249.2	84.3	271.5	1,506	1,553	3,059	333	206	392
August																		
Retained	526.8	1051.4	164.2	1.8	49.2	563.2	16.8	91.9	430.0	117.5	352.5	69.4	1,742	1,692	3,435	156	86	178
Released	282.9	509.0	48.3	4.2	32.8	254.4	2.6	47.4	235.7	129.9	117.5	104.1	840	929	1,769	261	133	293
TOTAL																		
Retained	1,431	3,037	598	6	237	1,545	103	257	1,704	570	777	702	5,066	5,901	10,967	238	167	290
Released	889	2,483	176	23	238	585	27	306	1,193	480	321	762	3,548	3,936	7,484	585	308	661
Catch	2,321	5,520	774	29	475	2,130	130	564	2,897	1,049	1,098	1,464	8,614	9,837	18,451	632	350	722

Table 20. Total (lodge + non-lodge) estimated Lingcod catch, by month and subarea, shown separately for released and retained fish, 2019.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	39.1	45.2	6.3	9.8	3.9	4.2	7.7	32.6	44.9	3.7	22.1	84	135	219	16	34	38
Released	10.6	6.4	7.4	1.1	5.9	3.9	0	0.5	2.5	9.0	0	0	24	23	47	8	9	12
June																		
Retained	71.0	514.9	0	0	31.8	33.5	4.1	33.1	305.4	129.3	23.7	54.6	586	615	1,201	107	82	135
Released	0	55.7	0	0	0	22.4	0	16.5	65.9	35.9	11.9	0	56	153	208	23	29	37
July																		
Retained	112.4	496.5	36.0	0	14.7	83.9	9.2	10.2	305.8	271.9	22.5	26.0	645	744	1,389	60	70	92
Released	4.6	0	0	0	0	34.5	4.0	0	149.9	90.6	0	0	5	279	284	3	62	62
August																		
Retained	327.6	542.8	4.5	0	23.4	54.5	5.2	27.3	126.3	457.7	130.1	4.6	875	829	1,704	161	98	188
Released	0	25.5	3.7	0	2.3	0	1.3	3.8	17.0	117.5	42.0	0	29	184	213	20	44	48
TOTAL																		
Retained	511	1,593	86	6	80	176	23	78	770	904	180	107	2,190	2,324	4,514	203	150	252
Released	15	88	11	1	8	61	5	21	235	253	54	0	114	638	752	32	82	88
Catch	526	1,681	97	7	88	237	28	99	1,005	1,157	234	107	2,304	2,962	5,266	205	171	267

Table 21. Total (lodge + non-lodge) estimated catch of other groundfish, by month and subarea, shown separately for released and retained fish, 2019. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	3.2	3.9	15.5	0	3.4	17.5	4.5	0	0	0	48	48	0	15	15
Released	21.1	12.7	14.7	0	0	50.4	2.8	5.8	25.0	0	3.7	0	49	88	136	12	16	20
June																		
Retained	12.5	111.4	0	0	21.2	29.8	0	8.3	26.9	21.5	26.4	0	124	134	258	53	26	59
Released	6.2	41.8	0	0.4	21.2	309.2	34.6	0	47.9	107.7	36.9	0	48	558	606	18	151	152
July																		
Retained	13.8	10.2	0	0	47.2	98.7	19.8	0	84.0	9.1	11.2	16.3	24	286	310	9	53	53
Released	4.6	0	0	2.7	17.7	207.3	60.6	0	75.0	68.0	14.1	1.6	5	447	451	3	53	53
August																		
Retained	28.3	114.5	0	0.6	11.7	115.1	0	4.3	21.9	43.3	50.4	1.2	143	248	391	65	53	84
Released	113.2	330.9	26.0	3.0	9.4	230.1	5.2	12.0	51.0	61.9	138.5	0	470	511	981	154	86	177
TOTAL																		
Retained	55	236	0	4	84	259	20	16	150	78	88	17	291	717	1,007	84	81	117
Released	145	385	41	6	48	797	103	18	199	238	193	2	571	1,603	2,175	156	182	240
Catch	200	621	41	10	132	1,056	123	34	349	316	281	19	862	2,320	3,182	177	199	267

Table 22. Total (lodge + non-lodge) estimated rockfish catch, by month and subarea, shown separately for released and retained fish, 2019. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	28.8	17.4	20.1	4.2	29.5	116.2	41.1	33.3	100.2	13.5	3.7	7.4	66	349	415	8	46	46
Released	31.7	19.1	22.1	11.6	19.6	50.4	60.9	42.4	112.7	26.9	22.1	36.8	73	383	456	25	52	58
June																		
Retained	136.8	262.7	52.2	0	38.9	104.3	52.9	57.9	284.4	193.9	6.6	45.5	452	784	1,236	81	102	130
Released	224.7	320.2	0	0	106.0	186.3	42.7	82.7	467.0	172.4	25.0	27.3	545	1,109	1,654	125	133	183
July																		
Retained	212.8	394.4	0	0.8	59.0	197.4	73.8	101.9	461.7	276.4	29.5	34.1	607	1,235	1,842	58	94	111
Released	152.1	71.3	8.1	0.4	185.8	345.4	81.7	40.8	776.6	543.7	5.6	16.3	231	1,996	2,228	52	174	182
August																		
Retained	268.8	783.0	40.8	0	46.9	199.8	11.6	41.2	187.1	433.0	272.8	9.3	1,093	1,202	2,294	141	126	189
Released	537.6	839.9	81.7	0	222.7	569.3	65.7	92.9	347.4	977.3	398.7	0	1,459	2,674	4,133	331	301	447
TOTAL																		
Retained	647	1,458	113	5	174	618	179	234	1,033	917	313	96	2,218	3,569	5,787	173	193	260
Released	946	1,250	112	12	534	1,151	251	259	1,704	1,720	451	80	2,308	6,163	8,472	358	376	519
Catch	1,593	2,708	225	17	708	1,769	430	493	2,737	2,637	764	177	4,526	9,733	14,259	398	423	581

Table 23. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species, 2019. Subareas treated as replicates.

Species	Kruskal-Wallis Statistics			Median Monthly Harvest			
	χ^2	df	P	May	June	July	August
Chinook Salmon	3.5	3	0.32	125.3	269.4	153.8	103.5
Chum Salmon	16.3	3	0.0010	0	0	0	0
Coho Salmon	24.8	3	< 0.0001	0	60.6	286.2	590.8
Pink Salmon	26.1	3	< 0.0001	0	0	107.1	47.4
Sockeye Salmon	3.0	3	0.39	0	0	0	0
Arrow Flounder	0.8	3	0.85	0	0	0	0
Big Skate	3.0	3	0.39	0	0	0	0
Cabezon	0.6	3	0.90	0	0	0	0
English Sole	3.7	3	0.29	0	0	0	0
Greenling	6.0	3	0.11	0	0	0	1.5
Lingcod	6.0	3	0.11	8.8	33.3	27.4	39.7
Pacific Cod	5.0	3	0.17	0	10.0	5.3	2.1
Pacific Halibut	6.9	3	0.07	53.3	169.3	205.8	128.3
Ratfish	0.0	3	1	0	0	0	0
Rock Sole	2.0	3	0.58	0	0	0	0
Sablefish	3.0	3	0.39	0	0	0	0
Starry Flounder	6.3	3	0.10	0	0	0	0
Black Rockfish	5.6	3	0.13	2.2	0	13.4	20.8
Boccacio Rockfish	0.0	3	1	0	0	0	0
Canary Rockfish	1.0	3	0.79	0	0	0	0
China Rockfish	7.3	3	0.06	0	0	0.8	0.5
Copper Rockfish	7.6	3	0.05	0	0.7	1.5	2.9
Dusky Rockfish	0.0	3	1	0	0	0	0
Quillback Rockfish	6.8	3	0.08	4.2	15.4	24.5	20.3
Redband Rockfish	3.0	3	0.39	0	0	0	0
Redstripe Rockfish	3.0	3	0.39	0	0	0	0
Tiger Rockfish	3.8	3	0.29	0	0	0	0
Widow Rockfish	0.0	3	1	0	0	0	0
Yelloweye Rockfish	0.0	3	1	0	0	0	0
Yellowtail Rockfish	0.4	3	0.94	0	0	0	0

Rockfish harvest was estimated at 5,787 fish (SE = 260; Table 22), composed mostly of unknown (33%), Black (*S. melanops*, 30%), Quillback (*Sebastes maliger*, 24%), Copper (*S. caurinus*, 6%), China (*S. nebulosus*, 4%), and five other rockfish species. In all, 8,472 (SE = 519) were released. Of the 791 Yelloweye Rockfish (*S. ruberrimus*) caught, all (100%) were released.

Biosampling

During interviews, observers were told about 3,027 retained Chinook Salmon, and were permitted to biosample 1,074 fish. The observed sex ratio was 53:47 (male:female, n = 270), and most fish were red-fleshed (89%, n = 612; Table 24). The median fork length of biosampled Chinook Salmon (n = 879) was 76 cm (95% of observations were between 52 and 100 cm; Figure 14). In all, 93 adipose-clipped individuals were encountered (Table 24), including 54 from standardized sampling methods (anglers were asked if their catch could be biosampled before any other interactions took place; Appendix 5) and 39 from non-standard interactions. Of

Table 24. Biosampling results, 2019.

	<i>Chinook Salmon</i>		<i>Coho Salmon</i>		<i>Pacific Halibut</i>	
	<i>Number</i>	<i>Percent</i>	<i>Number</i>	<i>Percent</i>	<i>Number</i>	<i>Percent</i>
<i>Number Examined</i>	1074		927		714	
<i>Subarea</i>						
3I	81	8%	86	9%	37	5%
3J	72	7%	86	9%	67	9%
3K	34	3%	66	7%	1	0%
4A	11	1%				
4B	175	16%	117	13%	29	4%
4C	149	14%	48	5%	147	21%
4D	80	7%	58	6%	23	3%
4E	4	0%			9	1%
4F	355	33%	340	37%	156	22%
4G	20	2%	28	3%	34	5%
4H	82	8%	73	8%	150	21%
4L	9	1%	17	2%	58	8%
<i>Sex</i>						
Male	142	53%	47	45%		
Female	128	47%	58	55%		
Undeterminable	386		509		523	
Unknown	418		313		191	
<i>Trip Type</i>						
Guided	417	39%	368	40%	254	36%
Unguided	657	61%	559	60%	460	64%
<i>Adipose Clips</i>						
Clipped	93	see	8	see		
Unmarked	958	App 5	913	App 5		
Unknown	9		4			
<i>Head Tagged</i>						
Yes	59	5%	5	1%		
No	1015	95%	922	99%		
<i>Scale Samples</i>						
Yes	938	87%	795	86%		
No	136	13%	132	14%		
<i>Flesh Colour</i>						
Red	552	89%				
White	60	10%				
Marble	9	1%				
Unknown						

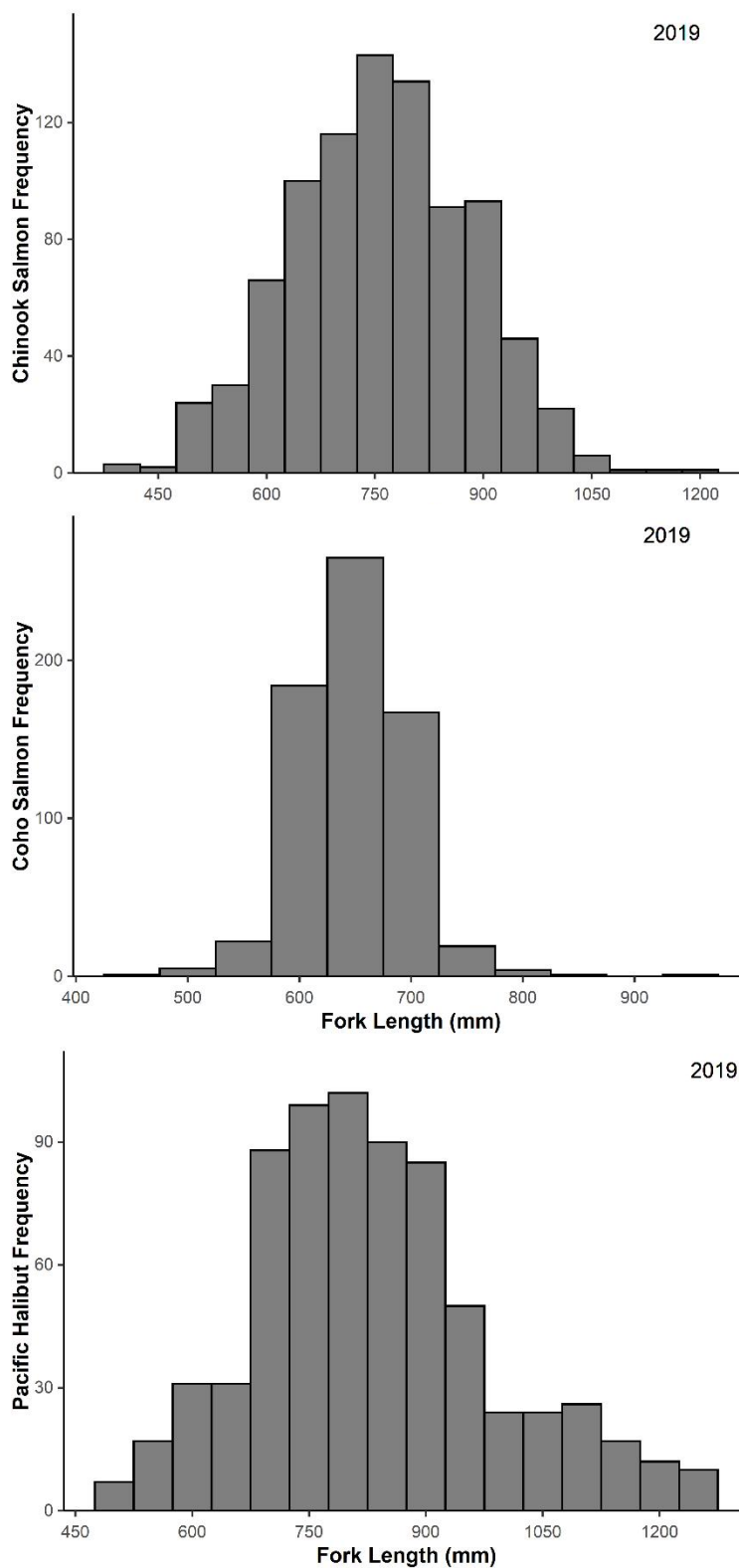


Figure 14. Length-frequency distributions of biosampled Chinook (n = 879) and Coho (n = 669) salmon, and Pacific Halibut (n = 713), 2019. Note that X and Y axis scales vary between plots.

Table 25. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2019.

	Stat Area 3				Stat Area 4			
	Net Weight		Round Weight		Net Weight		Round Weight	
	kg	lb	kg	lb	kg	lb	kg	lb
<i>Creel Interview (Non-Lodge) Average Weights</i>								
<i>Guided</i>	6.8	15.0	9.0	19.9	6.2	13.6	8.2	18.1
<i>Unguided</i>	6.5	14.3	8.6	18.9	5.6	12.4	7.4	16.4
<i>Overall</i>	6.6	14.6	8.8	19.3	5.8	12.8	7.7	17.0
<i>Estimated Non-lodge Harvest (number of fish)</i>								
<i>Overall</i>		4,221					5,901	
<i>Estimated Lodge Harvest (number of fish)</i>								
<i>Overall</i>		845					0	
<i>Biomass Harvested</i>								
<i>Non-lodge</i>	27,944	61,605	37,008	81,588	34,269	75,550	45,385	100,056
<i>Lodge</i>	5,594	12,332	7,408	16,332	0	0	0	0
<i>Overall</i>	33,537	73,937	44,416	97,920	34,269	75,550	45,385	100,056

the 93, 63% had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 24). Of the standardized samples, 8% were adipose-clipped (Appendix 5). Scale samples were obtained from 87% of the biosampled Chinook Salmon (Table 24).

For Coho Salmon, observers were told about 4,472 retained fish, and were permitted to biosample 927 fish. The observed sex ratio was 45:55 (male:female; n = 105; Table 24). The median Coho Salmon fork length (n = 669) was 65 cm (95% of observations were between 55 and 74 cm; Figure 14). In all, 8 adipose-clipped individuals were encountered (Table 24), including 4 from standardized sampling methods, and 4 from non-standard interactions. Of the 8, 63% had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 24). Of the standardized samples, 0.7% were adipose-clipped (Appendix 5). Scale samples were obtained from 86% of the biosampled Coho Salmon (Table 24).

Of the 2,333 Pacific Halibut that anglers reported as retained, 714 were biosampled (Table 24). None were sexed, and 713 were measured. The median size was 82 cm (95% of observations were between 55 and 120 cm; Figure 14). Average net and round weights, calculated from average lengths using regression formulas, were 6.6 kg (14.6 lb) and 8.8 kg (19.3 lb) in Area 3, and 5.8 kg (12.8 lb) and 7.7 kg (17.0 lb) in Area 4, respectively (Table 25).

Harvested Halibut Biomass

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 67.8 tonnes (149,487 lb), including 33.5 t (73,937 lb) in Area 3 and 34.3 t (75,550 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 89.8 t (197,976 lb) including 44.4 t (97,920 lb) in Area 3 and 45.4 t (100,056 lb) in Area 4 (Table 25).

DISCUSSION

Creel Estimates

In general, recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 20 years (Tables 26 and 27). Using published creel data collected since 1995, we found 11 years (1995-96, 2001-02, 2008-09¹, and 2015-2019) for which effort data were collected in June, July and August (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018,2019), and we have therefore limited this meta analyses to those three months. Meta-analysis revealed (Figure 15) that fishing effort has increased steadily (by 290 trips per year on average) since 1995. Effort in 2019 was similar to other recent years (2015-2019; though not quite as high as in 2017), and fell directly on the long-term trendline, supporting the notion that effort gradually increases over time.

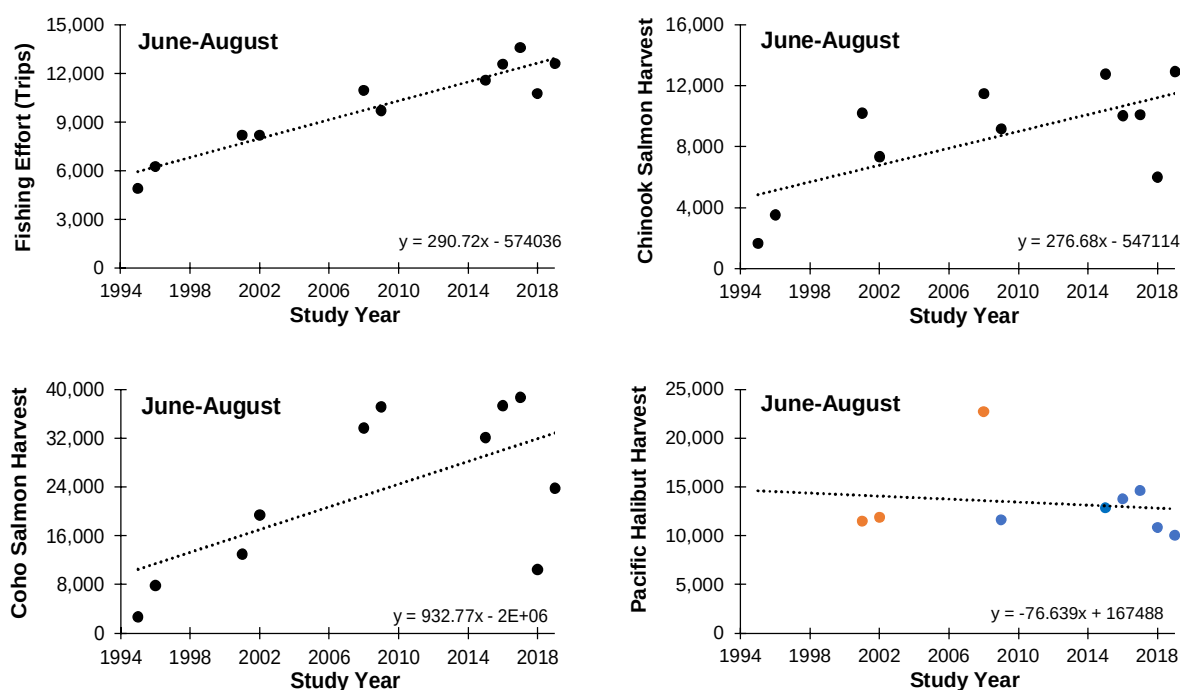


Figure 15. Fishing effort, and harvest of Chinook and Coho salmon, and Pacific Halibut from 1995 to 2019. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Linear regression equations are shown in the lower right of each panel. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).

¹ Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

Table 26. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2019, as estimated using creel survey methods.

Month	Year																						
	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
May	965	391	--	--	--	--	449	960	--	--	--	983	--	--	--	--	--	--	--	--	--	--	1,553
June	1,590	2,490	--	--	--	--	3,801	3,463	--	--	--	4,696	3,043	--	--	--	--	--	3,424	3,961	4,574	2,873	3,790
July	1,394	2,385	--	1,307	--	1,842	2,588	2,892	--	--	--	3,750	3,804	--	--	--	--	--	4,279	5,295	6,044	4,593	4,644
Aug	1,927	1,400	--	1,351	--	1,327	1,808	1,843	--	2,593	--	2,529	2,865	--	--	--	--	--	3,881	3,325	2,980	3,304	4,175
Sept	864	488	--	504	--	954	540	676	--	758	--	394	817	--	--	--	--	--	--	--	--	--	--
TOTAL	6,740	7,154	--	3,162	--	4,123	9,186	9,834	--	3,351	--	12,352	10,529	--	--	--	--	--	11,585	12,581	13,598	10,770	14,163

Table 27. Monthly Chinook Salmon, Coho Salmon and Pacific Halibut harvest in the Area 3 and 4 recreational fishery from 1995 to 2019, as estimated using creel survey methods.

Month	Year																						
	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Chinook Salmon Harvest																							
May	141	45	--	--	--	--	509	522	--	--	--	474	--	--	--	--	--	--	--	--	--	--	2,234
June	907	2,165	--	--	--	--	6,630	4,028	--	--	--	7,708	4,797	--	--	--	--	--	7,071	7,071	5,229	1,273	5,675
July	505	1,097	--	254	--	1,611	3,260	2,822	--	--	--	3,080	3,873	--	--	--	--	--	4,016	4,016	3,800	2,964	4,806
Aug	245	257	--	35	--	108	299	474	--	1,207	--	688	503	--	--	--	--	--	1,669	1,669	1,078	1,593	2,437
Sept	15	62	--	6	--	93	61	59	--	203	--	20	4	--	--	--	--	--	--	--	--	--	--
TOTAL	1,813	3,626	--	295	--	1,812	10,759	7,905	--	1,410	--	11,970	9,177	--	--	--	--	--	12,756	12,757	10,108	5,997	15,152
Coho Salmon Harvest																							
May	0	0	--	--	--	--	0	0	--	--	--	0	--	--	--	--	--	--	--	--	--	--	29
June	335	474	--	--	--	--	1,771	1,332	--	--	--	3,913	3,686	--	--	--	--	--	4,436	4,436	1,839	126	1,475
July	402	3,565	--	1,937	--	568	4,736	9,798	--	--	--	15,982	17,043	--	--	--	--	--	15,101	15,101	22,871	5,758	10,341
Aug	1,907	3,771	--	1,500	--	2,125	6,488	8,292	--	15,647	--	13,791	16,459	--	--	--	--	--	12,612	12,612	14,004	4,580	11,974
Sept	1,115	1,417	--	79	--	790	1,006	1,010	--	3,215	--	733	3,136	--	--	--	--	--	--	--	--	--	--
TOTAL	3,759	9,227	--	3,516	--	3,483	14,001	20,432	--	18,862	--	34,419	40,324	--	--	--	--	--	32,148	32,149	38,713	10,464	23,820
Pacific Halibut Harvest																							
May	--	--	--	--	--	--	281	467	--	--	--	1,435	--	--	--	--	--	--	--	--	--	--	941
June	--	--	--	--	--	--	4,872	3,912	--	--	--	8,367	3,029	--	--	--	--	--	3,551	3,551	4,632	3,407	2,698
July	--	--	--	--	--	3,963	3,535	4,716	--	--	--	8,507	4,466	--	--	--	--	--	4,096	4,096	6,563	4,263	3,894
Aug	--	--	--	1,801	--	2,530	3,085	3,247	--	8,178	--	5,865	4,142	--	--	--	--	--	5,201	5,201	3,410	3,183	3,435
Sept	--	--	--	328	--	1,224	222	615	--	1,812	--	774	1,205	--	--	--	--	--	--	--	--	--	--
TOTAL	--	--	--	2,129	--	7,717	11,995	12,957	--	9,990	--	24,948	12,842	--	--	--	--	--	12,848	12,849	14,606	10,853	10,967

Over the same period, Chinook and Coho salmon harvests have increased by 277 and 933 fish per year, on average (Table 27, Figure 15). The 2019 Chinook Salmon harvest was higher than that expected from the 20-year average rate of harvest increase, whereas the Coho Salmon harvest was much lower than would have been expected given the long-term trend. The Coho Salmon harvest of 23,790 fish was more than double that of the previous year, and was 9 times that in 1995, yet it was nevertheless only about 66% of that in 2015-17.

The Lingcod harvest in 2018 and 2019 were higher than in other recent years (3,052 in 2015, 1,803 in 2016, 2,772 in 2017, 4,270 in 2018, and 4,295 in 2019). This, despite a coast-wide reduction in the Lingcod bag limit from three to two fish starting in May 2018.

Pacific Halibut harvest did not show a strong among-year trend (Figure 15). While fishing effort increased, the total harvest of Pacific Halibut in recent years (2009, 2015-2019) is very similar to that from 2001-2002; Table 27). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 15).

The meta-analysis results make two important points: 1) the effort and CPUE of certain species has increased tremendously over the past ~20 years, potentially endangering stocks if left unchecked; and 2) management decisions from the recent past appear to have been effective, indicating that effective tools are available to keep effort and CPUE increases from going unchecked.

April and May Effort

In our report for the 2018 study year (Robichaud and Addison 2019), we made the case that substantial angling activity was being missed by starting the creel survey in June each year. Surveys in 1995 and 1996 (Van Tongeren and Winther 2010) included May, which accounted for 14% and 5% of the May-August effort, and for 8% and 1% of the May-August Chinook harvest. Based on trailer count data, we estimated that May effort in 2018 was likely more than 2,000 boat trips (~20% of the May to August effort). In 2019, we did start the survey in May, and found that May effort (1,553 boat trips) accounted for 11% of the May-August total, and for 15% of the May-August total Chinook harvest. Much of the early season angling targets Chinook Salmon, and occurs during a period in which the Area 3 and 4 stock composition is unknown or poorly resolved. As such, the Department of Fisheries and Oceans Canada (DFO) expressed interest in a May creel survey because genetic samples collected from Chinook Salmon would help them resolve stock composition and help determine exploitation rates on early-run-timing Chinook Salmon stocks. In 2019, during the month of May, 269 Chinook Salmon were biosampled, including 230 scale samples, and 22 tagged heads.

Following the same logic, we used trailer counts in 2019 to estimate if April also included large numbers of otherwise unaccounted fishing trips. Our estimates showed that April effort was likely about 65% of that in May, or about 1,000 boat trips (or 7% of the April to August effort). We are not currently recommending that the Area 3 & 4 creel surveys be extended into April. However, it may be worth the effort to continue to monitor April trailers counts, and to monitor for changes in April effort levels (and hence potential fisheries impacts to Chinook Salmon and

Pacific Halibut) should resident effort shift away from the typical fishery peak in June and July to take advantage of earlier opportunities.

Precision of the Results

Overall, our level of survey effort, both for CPE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2019 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon and Pacific Halibut), for which harvest estimates were precise to within 4-5%. Pink Salmon, Lingcod, and rockfish estimates were precise to within 14%, 11%, and 9%, respectively. Chum and Sockeye salmon estimates were not precise to within 15, though the survey was not designed to meet any level of precision for these last species.

The precision of our harvest estimates depended largely on the variability in the CPE and effort data. Differing levels of angler expertise, and day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions mean that fishing trips have a wide range of outcomes. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset. Sampling error is another main source of estimation imprecision; and by meeting our precision goals we have demonstrated that we are sampling an adequate proportion of the population.

Accuracy of the Results

The accuracy of the results depends on the quality of data provided by the anglers to the interviewers. Each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 1,601 of the 3,027 Chinook Salmon (53%) and 2,081 of the 4,472 Coho Salmon (47%) reported as kept during angler interviews were recorded as observed by the interviewer. While not every fish can be observed², each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We

² Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, or boat launch traffic congestion causing tension, among other reasons.

know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g. vessels outside of the survey area). There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 26 and 27 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.
- We recommend that future surveys continue to run for four months (May to August) rather than three (June to August). May surveys will help document and quantify angling effort and harvest during that month. The Department of Fisheries and Oceans Canada (DFO) has expressed interest in a May creel survey because genetic samples collected from Chinook Salmon will help resolve the currently unknown early-season Area 3 and 4 stock composition, which will help determine exploitation rates on early-run-timing Chinook Salmon stocks.
- Trailer counts should be conducted in April 2020 for comparison with April trailer counts in 2019 and continued throughout the creel survey season as in recent years.
- This report should be published in the Canadian Technical Report of Fisheries and Aquatic Sciences series.

ACKNOWLEDGEMENTS

We would like to thank the creel surveyors (Katelyn Cooper, Kathryn Bond, Anke De Wit, Stephen Fodor, and Gen Bock) for collecting large amounts of data and biosamples from anglers. Thanks to Pacific Salmon Commission, who provided support for this work through the Northern Fund. Thanks to Darren Chow and Shaun Davies for acting as liaison between our team and DFO. Thanks also go to the administrative staff at LGL and NCSFNSS, whose work made it possible for this study to be conducted. We also would like to thank Karl English for providing useful comments and edits on previous drafts of this report. Last but not least, we thank the many anglers, charter operators, and fishing lodges who participated in the survey.

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APPENDICES

APPENDIX 1: DATA FORMS.

2019 AREA 3/4 CREEL SURVEY
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE: _____ FLIGHT#: _____

SURVEYOR: _____

Note: only survey shaded subareas when conditions permit

Sub Area	RIGHT FRONT COUNT				TOTAL COUNT				FLIGHT TIME		SEA STATE (1-4)
	SPORT (Fishing)	SPORT (Moving)	COMM. (Fishing)	COMM. (Moving)	S (F)	S (M)	C (F)	C (M)	START	FINISH	
A											
C											
B											
C											
D											
E											
C											
F											
E											
F											
G											
L											
G											
J											
I											
K											
I											
H											
C											
A											

WEATHER / SEA STATE / COMMENTS:

SIGNATURE: _____

SHIFT SUMMARY SHEET

Landing Site: _____

Interviewer Name: _____

Interviewer Code: _____

Date: _____

Shift Code: _____

Shift Times: _____

Time Blocks	
From - To	TB
Before 7:00	1
7:00 - 7:59	2
8:00 - 8:59	3
9:00 - 9:59	4
10:00 - 10:59	5
11:00 - 11:59	6
12:00 - 12:59	7
13:00 - 13:59	8
14:00 - 14:59	9
15:00 - 15:59	10
16:00 - 16:59	11
17:00 - 17:59	12
18:00 - 18:59	13
19:00 - 19:59	14
20:00 - 20:59	15
21:00 - 21:59	16

* Fill in using numbers only.

Time Block	Fishing Conditions	Boats Missed	Not Fishing	Fishing

Totals:

--	--	--

Comments: _____

Trailer Count: _____ Time: _____

Database Tally # :

(office use only)

Site# _____ Obs# _____ Date / / Interview# _____
 DD/MM/YY

Pick ONE

☐ Refusal ☐ Partial

☐ Catch data given by angler (Catch not verified)

☐ Fish observed (Catch & Species verified)

☐ Adipose checked by interviewer

☐ Fish biosampled (e.g., length, sex, scales, etc.)

Pick ONE Target

☐	Shellfish Only
☐	Finfish Only
☐	Finfish and Shellfish

[illegible][illegible][illegible]

Weather Conditions:	

Comments: _____

Signature: _____

APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER BOAT TRIP) BY MONTH, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon harvested per boat trip in May 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	2.60	2.60	2.60	0.03	2.43	3.32	2.40	2.95	3.31	1.19	0.25	1.80
Unguided	2.17	1.44	1.44	0.09	1.02	0.97	1.23	1.16	1.15	0.20	0.29	0.40
Combined	3.25	1.86	1.86	0.08	1.12	1.37	1.44	1.62	1.75	1.13	0.27	2.00
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided												
Unguided												0.20
Combined												0.67
Pink Salmon												
Guided												
Unguided												
Combined												
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in June 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	4.00	1.78	1.50	0.02	0.45	3.08	3.70	2.66	2.49	0.18	2.55	1.92
Unguided	2.25	1.92	1.29	0.03	0.85	0.35	1.40	0.17	1.50	1.62	0.75	0.88
Combined	3.19	1.84	1.38	0.02	0.99	0.57	1.86	0.17	1.82	0.96	1.51	1.50
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided	1.61	0.33			0.21	2.17	0.20	0.75	0.84		0.35	0.67
Unguided	0.21	0.62			0.16	0.06	0.30		0.70	0.69	0.24	0.33
Combined	0.96	0.45			0.16	0.22	0.28		0.74	0.38	0.28	0.63
Pink Salmon												
Guided		0.06			0.01	0.08						
Unguided					0.03							
Combined		0.03			0.03	0.01						
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in July 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	1.98	1.79	2.65		1.40	1.00	1.56	2.09	2.16	0.64	1.02	0.21
Unguided	1.00	1.11	1.00	0.03	0.41	0.14	0.36	0.67	0.84	0.36	0.33	
Combined	1.47	1.59	2.00	0.02	0.45	0.22	0.51	0.57	1.29	0.47	0.63	0.17
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided	4.13	5.79	6.90		2.60	1.69	2.78	2.31	2.29	1.68	2.89	2.36
Unguided	1.29	3.00	2.23		0.54	0.22	0.51	0.33	1.50	1.67	1.35	1.14
Combined	2.67	4.98	5.06		0.63	0.35	0.80	0.29	1.77	1.67	2.02	2.11
Pink Salmon												
Guided	0.76	1.30	2.00		0.80	0.54	1.67	0.66	0.58	0.41	0.59	
Unguided	0.60	0.47	0.85		0.16	0.19	0.41		0.31	0.53	0.19	
Combined	0.68	1.06	1.55		0.19	0.22	0.57		0.40	0.48	0.37	
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in August 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	1.71	0.45	1.68		1.70	0.77	1.78	1.78	1.83	0.17	0.47	
Unguided	0.76	0.08	0.68		0.23	0.24	0.25	0.52	0.62	0.15		0.06
Combined	1.11	0.30	1.27		0.36	0.30	0.27	0.87	1.02	0.15	0.14	
Chum Salmon												
Guided							0.03	0.03	0.03			
Unguided	0.03		0.21					0.01	0.01		0.03	0.01
Combined	0.02		0.08					0.01	0.01		0.02	
Coho Salmon												
Guided	5.12	2.85	7.26		8.50	4.31	5.58	5.58	5.61	0.50	2.24	
Unguided	3.79	3.00	4.35	0.01	2.24	0.87	1.90	2.39	2.58	0.89	0.82	0.84
Combined	4.28	2.91	6.08	0.01	2.76	1.26	2.08	3.27	3.58	0.82	1.25	
Pink Salmon												
Guided	1.18	0.50	0.50		1.10	0.15	0.45	0.45	0.45		0.06	
Unguided	0.31	0.08	0.74	0.01	0.24	0.08	0.04	0.23	0.30	0.11	0.03	0.06
Combined	0.63	0.33	0.60	0.01	0.31	0.09	0.06	0.29	0.35	0.09	0.04	
Sockeye Salmon												
Guided												
Unguided	0.07											
Combined	0.04											
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in May 2019, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided						0.01						
Combined						0.01						
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided										0.20		0.05
Combined										0.13		
English Sole												
Guided												
Unguided												
Combined												
Greenling												
Guided												
Unguided				0.03	0.02							
Combined				0.02	0.02							
Lingcod												
Guided	0.60	0.60	0.60	0.20		0.05		0.18	0.25	0.69		0.60
Unguided					0.05		0.14	0.19	0.22	0.40	0.14	0.20
Combined		0.21	0.21	0.05	0.05	0.01	0.11	0.19	0.23	1.25	0.09	0.50
Pacific Cod												
Guided						0.05						
Unguided						0.02		0.11	0.17			
Combined						0.03		0.08	0.12			
Pacific Halibut												
Guided	1.80	1.80	1.80	0.13	0.29	1.42	0.80	0.82	0.88	2.31	2.25	3.40
Unguided	0.67	0.67	0.67		0.21	0.22	0.41	0.56	0.66	1.80	1.07	1.30
Combined	0.50	1.07	1.07	0.03	0.21	0.42	0.48	0.63	0.72	1.38	1.50	3.17
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided												
Combined												
Pacific Halibut												
Guided	1.80	1.80	1.80	0.13	0.29	1.42	0.80	0.82	0.88	2.31	2.25	3.40
Unguided	0.67	0.67	0.67		0.21	0.22	0.41	0.56	0.66	1.80	1.07	1.30
Combined	0.50	1.07	1.07	0.03	0.21	0.42	0.48	0.63	0.72	1.38	1.50	3.17
Sablefish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in June 2019, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided		0.06										
Unguided												
Combined		0.03										
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided											0.03	
Unguided									0.02			
Combined									0.01		0.01	
English Sole												
Guided												
Unguided												
Combined												
Greenling												
Guided		0.06										
Unguided					0.02			0.17				
Combined		0.03			0.02			0.17				
Lingcod												
Guided	0.14	0.89			0.02	0.17	0.20	0.85	0.95	0.73	0.13	0.50
Unguided	0.17	0.77			0.09	0.05		0.67	0.32	0.77	0.24	0.32
Combined	0.15	0.84			0.09	0.06	0.04	0.67	0.52	0.75	0.19	0.38
Pacific Cod												
Guided	0.07	0.33						0.04	0.05	0.27		
Unguided					0.04	0.01			0.03		0.35	0.26
Combined	0.04	0.19			0.04	0.01			0.04	0.13	0.20	
Pacific Halibut												
Guided	0.54	1.72	0.50		0.13	1.08	0.40	1.15	1.27	1.55	1.95	2.17
Unguided	1.00	0.69			0.11	0.64	0.20	1.17	0.64	0.38	1.07	1.00
Combined	0.75	1.29	0.23		0.15	0.67	0.24	1.17	0.84	0.92	1.44	2.13
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided						0.02						
Combined						0.02						
Pacific Halibut												
Guided	0.54	1.72	0.50		0.13	1.08	0.40	1.15	1.27	1.55	1.95	2.17
Unguided	1.00	0.69			0.11	0.64	0.20	1.17	0.64	0.38	1.07	1.00
Combined	0.75	1.29	0.23		0.15	0.67	0.24	1.17	0.84	0.92	1.44	2.13
Sablefish												
Guided												
Unguided						0.01						
Combined						0.01						
Starry Flounder												
Guided												
Unguided						0.01						
Combined						0.01						
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in July 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided						0.01			0.08			
Combined						0.01			0.05			
Big Skate												
Guided												
Unguided					0.04							
Combined					0.04							
Cabezon												
Guided												
Unguided						0.01						
Combined						0.01						
English Sole												
Guided											0.09	
Unguided						0.01						
Combined						0.01					0.04	
Greenling												
Guided												
Unguided		0.05			0.02	0.01			0.02			
Combined		0.02			0.02	0.01			0.01			
Lingcod												
Guided	0.22	0.38	0.05			0.38		0.23	0.25	1.55	0.25	0.39
Unguided	0.02	0.21	0.08		0.04	0.09	0.11	0.17	0.40	0.72	0.09	0.71
Combined	0.12	0.33	0.06		0.04	0.11	0.10	0.14	0.35	1.03	0.16	0.46
Pacific Cod												
Guided	0.04					0.38	0.11	0.03	0.02	0.05	0.09	0.29
Unguided	0.02					0.03	0.03		0.02	0.03		0.29
Combined	0.03					0.06	0.04		0.02	0.03	0.04	0.29
Pacific Halibut												
Guided	1.42	1.38	0.95		2.60	1.69	1.56	1.51	1.48	1.32	1.82	3.43
Unguided	0.73	1.63	0.08		0.16	0.45	0.31	0.67	0.41	0.69	0.88	2.43
Combined	1.06	1.45	0.61		0.26	0.56	0.47	1.14	0.77	0.93	1.29	3.23
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided						0.02						
Combined						0.02						
Pacific Halibut												
Guided	1.42	1.38	0.95		2.60	1.69	1.56	1.51	1.48	1.32	1.82	3.43
Unguided	0.73	1.63	0.08		0.16	0.45	0.31	0.67	0.41	0.69	0.88	2.43
Combined	1.06	1.45	0.61		0.26	0.56	0.47	1.14	0.77	0.93	1.29	3.23
Sablefish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided					0.20							
Unguided							0.20		0.01			
Combined					0.01		0.17		0.00			
Unknown Groundfish												
Guided								0.02	0.02			
Unguided					0.07	0.01						
Combined					0.07	0.01			0.01			

Average number of groundfish harvested per boat trip in August 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided					0.01							
Combined					0.01							
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided		0.05										
Unguided												
Combined		0.03										
English Sole												
Guided												
Unguided						0.04						
Combined						0.04						
Greenling												
Guided		0.05										
Unguided					0.04	0.03		0.01	0.01	0.26	0.05	0.13
Combined		0.03			0.03	0.03		0.00	0.00	0.21	0.04	
Lingcod												
Guided	0.06	1.40			1.00	0.46	0.20	0.20	0.21	1.50	0.53	0.29
Unguided	0.62	0.54	0.03			0.03	0.08	0.22	0.26	2.41	0.56	1.30
Combined	0.41	1.06	0.01		0.08	0.08	0.08	0.21	0.24	2.24	0.55	0.27
Pacific Cod												
Guided	0.12	0.30					0.03	0.03	0.03		0.35	0.07
Unguided		0.08				0.12		0.03	0.03		0.10	0.06
Combined	0.04	0.21				0.11		0.03	0.03		0.18	0.07
Pacific Halibut												
Guided	0.59	2.55	0.54	0.03	1.00	1.38	1.45	1.45	1.42	1.17	2.47	4.00
Unguided	0.76	1.54	0.32		0.10	0.74	0.15	0.44	0.53	0.44	1.08	0.87
Combined	0.70	2.15	0.45	0.02	0.17	0.82	0.25	0.72	0.82	0.58	1.50	4.00
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided								0.01	0.01			
Combined								0.00	0.00			
Pacific Halibut												
Guided	0.59	2.55	0.54	0.03	1.00	1.38	1.45	1.45	1.42	1.17	2.47	4.00
Unguided	0.76	1.54	0.32		0.10	0.74	0.15	0.44	0.53	0.44	1.08	0.87
Combined	0.70	2.15	0.45	0.02	0.17	0.82	0.25	0.72	0.82	0.58	1.50	4.00
Sablefish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of rockfish harvested per boat trip in May 2019, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided								0.05	0.06	0.13		
Unguided				0.02	0.05	0.11	0.36	0.13				
Combined				0.02	0.05	0.09	0.30	0.10	0.02	0.25		
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided												
Unguided				0.01								
Combined				0.01								
Copper Rockfish												
Guided												
Unguided					0.02							
Combined					0.02							
Dusky Rockfish												
Guided												
Unguided												
Combined												
Quillback Rockfish												
Guided				0.03		0.32	0.40	0.27	0.25	0.19	0.25	0.20
Unguided					0.02	0.01	0.32	0.11		0.20		0.05
Combined				0.01	0.02	0.06	0.33	0.15	0.07	0.13	0.09	0.17
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided								0.18	0.25			
Unguided												
Combined								0.05	0.07			
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided												
Unguided							0.05	0.02				
Combined							0.04	0.01				
Unknown rockfish												
Guided							1.80	0.77	0.50			
Unguided	0.17	0.11	0.11		0.06	0.14	0.09	0.39	0.56			
Combined	0.13	0.07	0.07		0.06	0.12	0.41	0.49	0.54			

Average number of rockfish harvested per boat trip in June 2019, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.07							0.12	0.14	0.27		
Unguided	0.17				0.01		0.08		0.08	0.31		0.06
Combined	0.12				0.01		0.06		0.10	0.29		
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided		0.06										
Unguided												
Combined		0.03										
China Rockfish												
Guided								0.05	0.06			
Unguided						0.01		0.33				
Combined						0.01		0.33	0.02			
Copper Rockfish												
Guided	0.07	0.06			0.01	0.08				0.18		
Unguided		0.23				0.01			0.04		0.02	0.01
Combined	0.04	0.13				0.01			0.03	0.08	0.01	
Dusky Rockfish												
Guided												
Unguided												
Combined												
Quillback Rockfish												
Guided	0.18	0.11					0.20	0.12	0.11	0.27	0.03	
Unguided	0.04				0.04	0.07	0.13	0.33	0.06		0.02	0.01
Combined	0.12	0.06			0.04	0.07	0.14	0.33	0.08	0.13	0.02	
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided								0.03	0.03			
Unguided						0.01						
Combined						0.01			0.01			
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided								0.01	0.02			
Unguided					0.01							
Combined					0.01				0.01			
Unknown rockfish												
Guided		0.33	0.17		0.09	0.92		0.33	0.38			0.42
Unguided	0.17				0.05	0.02	0.40	0.50	0.19	1.15	0.04	0.24
Combined	0.08	0.19	0.08		0.05	0.09	0.32	0.50	0.25	0.63	0.02	0.31

Average number of rockfish harvested per boat trip in July 2019, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.04	0.06				0.15		0.08	0.09	1.32	0.07	0.11
Unguided						0.02	0.18	0.33	0.17	0.17	0.05	
Combined	0.02	0.05				0.03	0.16	0.29	0.14	0.60	0.06	0.09
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided										0.06		
Combined										0.03		
China Rockfish												
Guided		0.02						0.03	0.03			
Unguided		0.05			0.02		0.02	0.17	0.02	0.06		
Combined		0.03			0.02		0.01	0.14	0.02	0.03		
Copper Rockfish												
Guided		0.04								0.14		
Unguided					0.01		0.03	0.17	0.01			
Combined		0.03			0.01		0.03	0.14	0.00	0.05		
Dusky Rockfish												
Guided												
Unguided												
Combined												
Quillback Rockfish												
Guided	0.04	0.02			0.20	0.23		0.15	0.17	0.45	0.16	0.04
Unguided	0.06	0.11		0.02	0.03	0.04	0.26	0.83	0.16	0.17	0.05	
Combined	0.05	0.05		0.01	0.04	0.06	0.23	0.71	0.16	0.28	0.10	0.03
Redband Rockfish												
Guided												
Unguided									0.01			
Combined									0.00			
Redstripe Rockfish												
Guided												
Unguided									0.01			
Combined									0.00			
Tiger Rockfish												
Guided								0.01	0.01	0.05		
Unguided									0.01			
Combined									0.01	0.02		
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided												
Unguided						0.03					0.02	
Combined						0.03					0.01	
Unknown rockfish												
Guided	0.22	0.19				0.15				0.05		0.61
Unguided	0.10	0.11			0.10	0.14	0.43	0.17	0.28	0.03	0.07	
Combined	0.16	0.17			0.10	0.14	0.37	0.14	0.18	0.03	0.04	0.49

Average number of rockfish harvested per boat trip in August 2019, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.18	0.75				0.15	0.05	0.05	0.06		0.12	
Unguided	0.31				0.02	0.05	0.02	0.06	0.07	1.93	0.62	1.13
Combined	0.26	0.45			0.02	0.06	0.02	0.06	0.07	1.58	0.46	
Bocaccio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided		0.05										
Unguided												
Combined		0.03										
China Rockfish												
Guided		0.10	0.04									
Unguided								0.01	0.01	0.41	0.13	0.24
Combined		0.06	0.02					0.01	0.01	0.33	0.09	
Copper Rockfish												
Guided		0.15	0.06			0.23	0.01	0.01	0.01			
Unguided	0.03							0.01	0.01		0.18	0.10
Combined	0.02	0.09	0.04			0.03		0.01	0.01		0.13	
Dusky Rockfish												
Guided												
Unguided												
Combined												
Quillback Rockfish												
Guided	0.12	0.55	0.12			0.08	0.09	0.09	0.10		0.65	
Unguided	0.14	0.23			0.02	0.11		0.06	0.08	0.11	0.28	0.21
Combined	0.13	0.42	0.07		0.02	0.11		0.07	0.08	0.09	0.39	
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided		0.05					0.03	0.03	0.03			
Unguided						0.01		0.01	0.01		0.08	0.04
Combined		0.03				0.01		0.01	0.02		0.05	
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided							0.01	0.01	0.01			
Unguided							0.02	0.01	0.01			
Combined							0.02	0.01	0.01			
Unknown rockfish												
Guided		0.05			0.60	0.23	0.09	0.09	0.10	0.33		0.57
Unguided		0.23			0.09	0.07	0.15	0.18	0.19	0.07	0.05	0.06
Combined		0.12			0.13	0.09	0.14	0.16	0.16	0.12	0.04	0.53

APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2019, along with numbers that were estimated to be lodge boats. Weekend and holiday flights are shaded.

Month	Date	Subarea												Area 3
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Lodge Boats
May	1 May 2019	0	0	0	2	8	2	0	1	0	0	0	0	0
	5 May 2019	2	3	1	6	1	14	1	0	2	0	0	0	0
	11 May 2019	0	0	3	2	7	8	0	0	0	1	2	0	0
	14 May 2019	3	1	2	0	4	2	0	0	1	0	0	0	4
	17 May 2019	2	4	2	0	0	6	2	4	2	1	1	0	0
	24 May 2019	1	1	2	3	2	5	0	0	0	0	0	0	0
	26 May 2019	4	1	5	5	5	15	3	0	9	1	3	5	6
	31 May 2019	7	5	9	3	2	15	0	0	7	2	1	2	8
June	3 Jun 2019	1	2	5	2	1	0	1	0	4	0	0	0	0
	9 Jun 2019	3	4	14	2	3	14	1	0	1	0	0	0	12
	12 Jun 2019	7	10	3	1	6	16	1	0	9	5	1	6	11
	15 Jun 2019	13	21	14	1	9	15	1	2	14	1	5	0	9
	18 Jun 2019	6	14	9	0	4	5	0	0	6	1	2	0	11
	24 Jun 2019	6	11	13	2	6	6	0	0	12	1	5	3	4
	28 Jun 2019	6	6	5	2	8	13	3	0	16	3	2	7	4
	30 Jun 2019	21	20	9	1	14	13	8	6	19	14	3	3	12
July	1 Jul 2019	13	14	2	5	14	17	2	2	11	5	2	0	4
	3 Jul 2019	7	6	6	2	2	15	1	1	23	1	6	2	8
	8 Jul 2019	0	3	5	2	1	11	1	2	20	5	2	0	4
	16 Jul 2019	8	30	10	1	9	15	0	1	17	3	1	0	10
	13 Jul 2019	14	25	3	2	4	15	2	3	28	10	8	4	12
	21 Jul 2019	15	39	8	3	8	22	8	3	12	14	1	4	12
	26 Jul 2019	10	11	5	0	12	8	1	0	12	1	2	0	0
	30 Jul 2019	17	9	12	1	6	13	1	0	10	5	0	0	8
August	1 Aug 2019	16	12	7	4	13	9	2	2	13	10	9	0	8
	3 Aug 2019	39	15	6	2	10	30	2	5	9	4	10	1	12
	8 Aug 2019	17	16	13	4	5	19	5	1	22	2	5	1	8
	12 Aug 2019	16	19	2	0	1	11	1	3	20	6	7	0	8
	18 Aug 2019	6	4	16	2	3	14	0	1	2	1	0	0	11
	26 Aug 2019	5	1	4	1	5	2	0	1	4	0	1	0	4
	28 Aug 2019	15	7	7	3	4	17	0	1	6	7	3	0	3
	31 Aug 2019	17	6	13	2	3	12	0	8	3	1	3	1	0

APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA AND SPECIES.

Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	274.3	165.7	191.5	10.5	227.9	592.6	55.2	67.0	250.4	40.4	11.1	88.3	1,975
Lodge Report	112.5	67.9	78.6	0	0	0	0	0	0	0	0	0	259
MAY RETAINED	386.8	233.6	270.1	10.5	227.9	592.6	55.2	67.0	250.4	40.4	11.1	88.3	2,234
SE	80.8	20.5	15.6	2.7	21.2	63.5	10.6	1.6	32.4	17.4	5.9	22.0	117
June													
Creel Estimate	1036.3	793.6	475.1	1.8	356.8	320.4	189.1	8.3	1062.8	165.2	188.5	218.4	4,816
Lodge Report	386.2	295.8	177.1	0	0	0	0	0	0	0	0	0	859
JULY RETAINED	1422.4	1089.3	652.1	1.8	356.8	320.4	189.1	8.3	1062.8	165.2	188.5	218.4	5,675
SE	83.0	116.2	81.8	0.8	37.1	42.9	22.4	8.3	64.6	42.0	23.3	80.3	209
July													
Creel Estimate	631.4	1069.1	536.1	1.5	162.2	162.8	47.4	40.8	1133.4	122.3	89.9	9.8	4,007
Lodge Report	225.6	381.9	191.5	0	0	0	0	0	0	0	0	0	799
JULY RETAINED	856.9	1451.0	727.6	1.5	162.2	162.8	47.4	40.8	1133.4	122.3	89.9	9.8	4,806
SE	56.7	108.4	85.6	0.7	21.4	28.0	9.3	18.8	71.0	31.4	12.0	3.4	174
August													
Creel Estimate	721.5	127.3	397.3	0	100.8	205.9	18.0	111.6	532.1	30.9	33.6	0	2,279
Lodge Report	91.5	16.1	50.4	0	0	0	0	0	0	0	0	0	158
AUGUST RETAINED	813.0	143.4	447.7	0	100.8	205.9	18.0	111.6	532.1	30.9	33.6	0	2,437
SE	114.6	46.5	40.1	0	14.4	39.8	4.5	0.8	47.1	11.1	17.5	0	146
TOTAL													
Creel Estimate	2,663	2,156	1,600	14	848	1,282	310	228	2,979	359	323	316	13,077
Lodge Report	816	762	498	0	0	0	0	0	0	0	0	0	2,075
TOTAL RETAINED	3,479	2,917	2,098	14	848	1,282	310	228	2,979	359	323	316	15,152
SE	172	167	126	3	50	91	27	21	112	56	32	83	330

Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	232.1	140.2	162.1	10.5	94.3	220.8	9.9	23.2	102.7	18.0	1.8	36.8	1,052
Lodge Report	40.4	24.4	28.2	0	0	0	0	0	0	0	0	0	93
MAY RELEASED	272.5	164.6	190.3	10.5	94.3	220.8	9.9	23.2	102.7	18.0	1.8	36.8	1,145
SE	63.0	17.3	13.2	3.3	16.4	32.3	4.5	0.6	20.3	10.3	1.3	17.5	81
June													
Creel Estimate	992.6	612.6	897.4	1.1	219.0	346.5	85.4	0	625.7	114.9	235.9	200.2	4,331
Lodge Report	160.2	98.9	144.9	0	0	0	0	0	0	0	0	0	404
JULY RELEASED	1152.8	711.5	1042.2	1.1	219.0	346.5	85.4	0	625.7	114.9	235.9	200.2	4,735
SE	116.7	137.1	216.0	0.6	27.5	50.8	12.7	0	57.7	36.6	53.9	77.6	310
July													
Creel Estimate	258.1	366.5	544.2	1.1	106.2	202.3	60.6	10.2	437.8	45.3	33.7	8.1	2,074
Lodge Report	68.0	96.6	143.4	0	0	0	0	0	0	0	0	0	308
JULY RELEASED	326.1	463.1	687.6	1.1	106.2	202.3	60.6	10.2	437.8	45.3	33.7	8.1	2,382
SE	39.3	63.5	79.7	0.8	19.3	41.2	11.2	7.3	56.2	13.2	8.2	6.4	133
August													
Creel Estimate	763.9	623.6	505.0	0	117.2	169.6	12.9	78.1	371.7	61.9	67.1	0	2,771
Lodge Report	38.3	31.3	25.4	0	0	0	0	0	0	0	0	0	95
AUGUST RELEASED	802.3	654.9	530.4	0	117.2	169.6	12.9	78.1	371.7	61.9	67.1	0	2,866
SE	139.5	250.0	54.1	0	22.6	40.1	5.0	0.6	52.5	33.7	31.5	0	303
TOTAL													
Creel Estimate	2,247	1,743	2,109	13	537	939	169	111	1,538	240	339	245	10,229
Lodge Report	307	251	342	0	0	0	0	0	0	0	0	0	900
TOTAL RELEASED	2,554	1,994	2,450	13	537	939	169	111	1,538	240	339	245	11,129
SE	196	293	237	3	44	83	18	7	98	52	63	80	461

Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	29.4	29
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	29.4	29
SE	0	0	0	0	0	0	0	0	0	0	0	21.4	21
June													
Creel Estimate	312.1	194.9	0	0	56.5	126.7	28.5	0	434.1	64.6	35.6	91.0	1,344
Lodge Report	80.7	50.4	0	0	0	0	0	0	0	0	0	0	131
JULY RETAINED	392.8	245.3	0	0	56.5	126.7	28.5	0	434.1	64.6	35.6	91.0	1,475
SE	51.7	72.1	0	0	15.0	37.7	7.0	0	42.0	26.4	8.4	40.4	117
July													
Creel Estimate	1142.9	3349.7	1356.5	0	224.2	261.5	73.8	20.4	1553.2	439.5	286.6	120.3	8,829
Lodge Report	295.5	866.0	350.7	0	0	0	0	0	0	0	0	0	1,512
JULY RETAINED	1438.4	4215.7	1707.1	0	224.2	261.5	73.8	20.4	1553.2	439.5	286.6	120.3	10,341
SE	117.9	288.4	178.4	0	37.5	46.1	14.2	14.6	86.7	71.1	35.2	30.1	384
August													
Creel Estimate	2787.0	1221.7	1897.6	0.6	783.0	872.1	136.6	420.0	1868.4	167.0	293.8	0	10,448
Lodge Report	720.5	315.8	490.5	0	0	0	0	0	0	0	0	0	1,527
AUGUST RETAINED	3507.5	1537.5	2388.1	0.6	783.0	872.1	136.6	420.0	1868.4	167.0	293.8	0	11,974
SE	258.3	181.9	90.1	0.5	64.6	103.9	21.9	3.0	105.4	44.9	66.7	0	375
TOTAL													
Creel Estimate	4,242	4,766	3,254	1	1,064	1,260	239	440	3,856	671	616	241	20,650
Lodge Report	1,097	1,232	841	0	0	0	0	0	0	0	0	0	3,170
TOTAL RETAINED	5,339	5,998	4,095	1	1,064	1,260	239	440	3,856	671	616	241	23,820
SE	289	349	200	0	76	120	27	15	143	88	76	55	550

Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	44.1	44
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	44.1	44
SE	0	0	0	0	0	0	0	0	0	0	0	32.2	32
June													
Creel Estimate	31.2	111.4	0	0	7.1	33.5	14.2	0	56.9	14.4	2.6	0	271
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	31.2	111.4	0	0	7.1	33.5	14.2	0	56.9	14.4	2.6	0	271
SE	13.0	53.0	0	0	5.0	11.4	6.0	0	14.9	10.2	1.9	0	59
July													
Creel Estimate	161.3	203.6	138.1	0	38.3	19.7	9.2	0	164.9	0	59.0	14.6	809
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	161.3	203.6	138.1	0	38.3	19.7	9.2	0	164.9	0	59.0	14.6	809
SE	45.6	52.4	86.2	0	18.3	8.8	2.9	0	31.7	0	18.3	6.7	119
August													
Creel Estimate	367.8	63.6	63.1	0	103.2	157.5	5.2	34.0	162.8	0	25.2	0	982
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	367.8	63.6	63.1	0	103.2	157.5	5.2	34.0	162.8	0	25.2	0	982
SE	93.7	35.5	16.7	0	22.3	39.4	3.1	0.3	33.4	0	8.8	0	117
TOTAL													
Creel Estimate	560	379	201	0	149	211	29	34	385	14	87	59	2,107
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	560	379	201	0	149	211	29	34	385	14	87	59	2,107
SE	105	83	88	0	29	42	7	0	48	10	20	33	179

Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	13.9	0	0	10.6	3.7	0	0	0	0	0	0	28
Lodge Report	0	0.5	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	14.4	0	0	10.6	3.7	0	0	0	0	0	0	29
SE	0	9.9	0	0	5.6	2.7	0	0	0	0	0	0	12
July													
Creel Estimate	290.3	712.7	414.3	0	67.8	167.8	52.7	0	350.8	126.9	52.0	0	2,235
Lodge Report	9.5	23.3	13.6	0	0	0	0	0	0	0	0	0	46
JULY RETAINED	299.9	736.1	427.8	0	67.8	167.8	52.7	0	350.8	126.9	52.0	0	2,282
SE	53.1	119.8	91.9	0	17.6	32.4	12.6	0	43.2	35.9	12.7	0	175
August													
Creel Estimate	410.3	140.0	185.7	0.6	89.1	60.6	3.9	37.4	182.2	18.6	8.4	0	1,137
Lodge Report	13.4	4.6	6.1	0	0	0	0	0	0	0	0	0	24
AUGUST RETAINED	423.7	144.6	191.8	0.6	89.1	60.6	3.9	37.4	182.2	18.6	8.4	0	1,161
SE	143.2	100.2	34.9	0.5	19.5	15.8	1.9	0.3	29.9	11.8	4.5	0	183
TOTAL													
Creel Estimate	701	867	600	1	168	232	57	37	533	145	60	0	3,400
Lodge Report	23	28	20	0	0	0	0	0	0	0	0	0	71
TOTAL RETAINED	724	895	620	1	168	232	57	37	533	145	60	0	3,471
SE	153	156	98	0	27	36	13	0	53	38	13	0	253

Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	14.4	0	0	14
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	14.4	0	0	14
SE	0	0	0	0	0	0	0	0	0	10.2	0	0	10
July													
Creel Estimate	705.1	1293.1	991.0	0	85.5	177.6	46.1	0	368.8	108.7	54.8	0	3,831
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	705.1	1293.1	991.0	0	85.5	177.6	46.1	0	368.8	108.7	54.8	0	3,831
SE	129.8	216.7	279.3	0	24.5	52.8	15.0	0	67.3	41.8	15.4	0	390
August													
Creel Estimate	396.1	407.2	382.5	0	164.1	30.3	24.5	43.6	174.9	12.4	4.2	0	1,640
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	396.1	407.2	382.5	0	164.1	30.3	24.5	43.6	174.9	12.4	4.2	0	1,640
SE	100.0	117.1	104.6	0	33.8	12.9	10.8	0.3	38.3	7.4	3.2	0	194
TOTAL													
Creel Estimate	1,101	1,700	1,373	0	250	208	71	44	544	135	59	0	5,485
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	1,101	1,700	1,373	0	250	208	71	44	544	135	59	0	5,485
SE	164	246	298	0	42	54	18	0	77	44	16	0	435

Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	14.1	0	26.0	0	0	0	0	1.4	7.3	0	4.2	0	53
Lodge Report	3.2	0.0	5.8	0	0	0	0	0	0	0	0	0	9
AUGUST RETAINED	17.3	0	31.8	0	0	0	0	1.4	7.3	0	4.2	0	62
SE	10.2	0	16.0	0	0	0	0	0.0	4.7	0	3.2	0	20
TOTAL													
Creel Estimate	14	0	26	0	0	0	0	1	7	0	4	0	53
Lodge Report	3	0	6	0	0	0	0	0	0	0	0	0	9
TOTAL RETAINED	17	0	32	0	0	0	0	1	7	0	4	0	62
SE	10	0	16	0	0	0	0	0	5	0	3	0	20

Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	0	0	3.7	0	0	0	0	0	0	0	8.4	0	12
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	3.7	0	0	0	0	0	0	0	8.4	0	12
SE	0	0	2.6	0	0	0	0	0	0	0	4.5	0	5
TOTAL													
Creel Estimate	0	0	4	0	0	0	0	0	0	0	8	0	12
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	4	0	0	0	0	0	0	0	8	0	12
SE	0	0	3	0	0	0	0	0	0	0	5	0	5

Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	28.3	0	0	0	0	0	0	0	0	0	0	0	28
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	28.3	0	0	0	0	0	0	0	0	0	0	0	28
SE	20.3	0	0	0	0	0	0	0	0	0	0	0	20
TOTAL													
Creel Estimate	28	0	0	0	0	0	0	0	0	0	0	0	28
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	28	0	0	0	0	0	0	0	0	0	0	0	28
SE	20	0	0	0	0	0	0	0	0	0	0	0	20

Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	1.3	0	0	0	0	0	1
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	1.3	0	0	0	0	0	1
SE	0	0	0	0	0	0	1.0	0	0	0	0	0	1
August													
Creel Estimate	0	0	0	0	2.3	6.1	0	1.0	4.9	0	0	0	14
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	2.3	6.1	0	1.0	4.9	0	0	0	14
SE	0	0	0	0	1.8	4.3	0	0.0	4.2	0	0	0	6
TOTAL													
Creel Estimate	0	0	0	0	2	6	1	1	5	0	0	0	16
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	2	6	1	1	5	0	0	0	16
SE	0	0	0	0	2	4	1	0	4	0	0	0	6

Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	42.2	95.6	110.5	4.2	43.2	182.1	18.4	26.0	102.7	49.4	60.9	139.7	875
Lodge Report	11.2	25.4	29.4	0	0	0	0	0	0	0	0	0	66
MAY RETAINED	53.4	121.0	139.9	4.2	43.2	182.1	18.4	26.0	102.7	49.4	60.9	139.7	941
SE	17.9	20.6	9.0	3.0	11.8	29.2	5.7	0.6	17.9	11.4	9.5	13.2	51
June													
Creel Estimate	243.5	556.9	79.2	0	53.0	380.0	24.4	57.9	491.0	158.0	180.6	309.4	2,534
Lodge Report	45.4	103.8	14.8	0	0	0	0	0	0	0	0	0	164
JULY RETAINED	288.9	660.7	93.9	0	53.0	380.0	24.4	57.9	491.0	158.0	180.6	309.4	2,698
SE	49.3	87.7	41.1	0	12.6	38.4	5.8	37.2	46.0	30.3	13.7	48.2	143
July													
Creel Estimate	456.3	977.4	162.5	0	91.4	419.4	43.5	81.5	680.6	244.7	182.7	183.7	3,524
Lodge Report	105.8	226.6	37.7	0	0	0	0	0	0	0	0	0	370
JULY RETAINED	562.0	1204.0	200.1	0	91.4	419.4	43.5	81.5	680.6	244.7	182.7	183.7	3,894
SE	49.6	121.3	47.1	0	19.9	53.1	8.4	28.3	63.1	38.6	16.6	10.9	171
August													
Creel Estimate	452.7	903.5	141.1	1.8	49.2	563.2	16.8	91.9	430.0	117.5	352.5	69.4	3,190
Lodge Report	74.1	147.8	23.1	0	0	0	0	0	0	0	0	0	245
AUGUST RETAINED	526.8	1051.4	164.2	1.8	49.2	563.2	16.8	91.9	430.0	117.5	352.5	69.4	3,435
SE	78.0	133.3	23.6	1.4	12.2	52.8	5.6	0.8	43.7	33.4	37.8	1.4	178
TOTAL													
Creel Estimate	1,195	2,533	493	6	237	1,545	103	257	1,704	570	777	702	10,122
Lodge Report	236	504	105	0	0	0	0	0	0	0	0	0	845
TOTAL RETAINED	1,431	3,037	598	6	237	1,545	103	257	1,704	570	777	702	10,967
SE	106	202	67	3	29	89	13	47	91	60	45	51	290

Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	10.6	38.2	44.2	18.9	3.9	38.7	14.2	12.1	37.6	35.9	12.9	58.8	326
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	10.6	38.2	44.2	18.9	3.9	38.7	14.2	12.1	37.6	35.9	12.9	58.8	326
SE	8.4	25.4	3.6	13.4	2.0	13.9	5.9	0.3	13.4	14.6	4.5	42.9	58
June													
Creel Estimate	93.6	988.5	26.4	0	113.0	149.0	2.0	33.1	425.1	64.6	106.8	327.6	2,330
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	93.6	988.5	26.4	0	113.0	149.0	2.0	33.1	425.1	64.6	106.8	327.6	2,330
SE	31.8	401.5	19.0	0	70.5	42.7	1.5	24.5	90.4	31.1	16.5	122.6	441
July													
Creel Estimate	502.3	946.9	56.9	0	88.5	143.1	7.9	214.0	494.7	249.2	84.3	271.5	3,059
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	502.3	946.9	56.9	0	88.5	143.1	7.9	214.0	494.7	249.2	84.3	271.5	3,059
SE	126.5	306.5	34.4	0	32.5	40.2	3.1	144.9	80.2	102.2	16.9	41.2	392
August													
Creel Estimate	282.9	509.0	48.3	4.2	32.8	254.4	2.6	47.4	235.7	129.9	117.5	104.1	1,769
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	282.9	509.0	48.3	4.2	32.8	254.4	2.6	47.4	235.7	129.9	117.5	104.1	1,769
SE	156.8	208.4	17.4	2.8	17.6	82.2	1.5	0.4	43.2	81.9	38.6	25.0	293
TOTAL													
Creel Estimate	889	2,483	176	23	238	585	27	306	1,193	480	321	762	7,484
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	889	2,483	176	23	238	585	27	306	1,193	480	321	762	7,484
SE	204	547	43	14	80	102	7	147	129	135	45	139	661

Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	19.1	22.1	6.3	9.8	3.9	4.2	7.7	32.6	44.9	3.7	22.1	176
Lodge Report	0	19.9	23.1	0	0	0	0	0	0	0	0	0	43
MAY RETAINED	0	39.1	45.2	6.3	9.8	3.9	4.2	7.7	32.6	44.9	3.7	22.1	219
SE	0	15.4	1.8	4.5	7.3	2.8	3.0	0.2	9.9	27.1	2.7	16.1	38
June													
Creel Estimate	49.9	362.0	0	0	31.8	33.5	4.1	33.1	305.4	129.3	23.7	54.6	1,027
Lodge Report	21.1	152.9	0	0	0	0	0	0	0	0	0	0	174
JULY RETAINED	71.0	514.9	0	0	31.8	33.5	4.1	33.1	305.4	129.3	23.7	54.6	1,201
SE	18.3	105.2	0	0	15.6	14.7	2.1	24.5	50.4	37.7	6.2	40.4	135
July													
Creel Estimate	50.7	224.0	16.2	0	14.7	83.9	9.2	10.2	305.8	271.9	22.5	26.0	1,035
Lodge Report	61.7	272.5	19.8	0	0	0	0	0	0	0	0	0	354
JULY RETAINED	112.4	496.5	36.0	0	14.7	83.9	9.2	10.2	305.8	271.9	22.5	26.0	1,389
SE	20.0	55.6	9.3	0	5.6	23.7	4.3	7.3	45.2	45.1	5.4	11.7	92
August													
Creel Estimate	268.8	445.4	3.7	0	23.4	54.5	5.2	27.3	126.3	457.7	130.1	4.6	1,547
Lodge Report	58.8	97.4	0.8	0	0	0	0	0	0	0	0	0	157
AUGUST RETAINED	327.6	542.8	4.5	0	23.4	54.5	5.2	27.3	126.3	457.7	130.1	4.6	1,704
SE	95.8	129.2	2.6	0	14.6	20.5	3.1	0.3	27.3	80.2	42.5	3.3	188
TOTAL													
Creel Estimate	369	1,050	42	6	80	176	23	78	770	904	180	107	3,786
Lodge Report	142	543	44	0	0	0	0	0	0	0	0	0	728
TOTAL RETAINED	511	1,593	86	6	80	176	23	78	770	904	180	107	4,514
SE	100	176	10	4	23	35	6	26	74	103	43	45	252

Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2019.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	10.6	6.4	7.4	1.1	5.9	3.9	0	0.5	2.5	9.0	0	0	47
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	10.6	6.4	7.4	1.1	5.9	3.9	0	0.5	2.5	9.0	0	0	47
SE	8.4	0.8	0.6	0.7	4.4	2.8	0	0.0	1.8	6.8	0	0	12
June													
Creel Estimate	0	55.7	0	0	0	22.4	0	16.5	65.9	35.9	11.9	0	208
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	55.7	0	0	0	22.4	0	16.5	65.9	35.9	11.9	0	208
SE	0	23.5	0	0	0	8.3	0	16.5	16.4	14.7	4.5	0	37
July													
Creel Estimate	4.6	0	0	0	0	34.5	4.0	0	149.9	90.6	0	0	284
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	4.6	0	0	0	0	34.5	4.0	0	149.9	90.6	0	0	284
SE	3.3	0	0	0	0	18.0	1.7	0	54.2	23.4	0	0	62
August													
Creel Estimate	0	25.5	3.7	0	2.3	0	1.3	3.8	17.0	117.5	42.0	0	213
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	25.5	3.7	0	2.3	0	1.3	3.8	17.0	117.5	42.0	0	213
SE	0	20.0	2.6	0	1.8	0	1.1	0.0	7.0	36.7	22.6	0	48
TOTAL													
Creel Estimate	15	88	11	1	8	61	5	21	235	253	54	0	752
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	15	88	11	1	8	61	5	21	235	253	54	0	752
SE	9	31	3	1	5	20	2	17	57	46	23	0	88

Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2019. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	3.2	3.9	15.5	0	3.4	17.5	4.5	0	0	48
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	3.2	3.9	15.5	0	3.4	17.5	4.5	0	0	48
SE	0	0	0	2.2	2.9	6.7	0	0.1	12.6	3.4	0	0	15
June													
Creel Estimate	12.5	111.4	0	0	21.2	29.8	0	8.3	26.9	21.5	26.4	0	258
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	12.5	111.4	0	0	21.2	29.8	0	8.3	26.9	21.5	26.4	0	258
SE	8.8	51.9	0	0	8.7	10.6	0	8.3	9.5	15.3	10.3	0	0
July													
Creel Estimate	13.8	10.2	0	0	47.2	98.7	19.8	0	84.0	9.1	11.2	16.3	310
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	13.8	10.2	0	0	47.2	98.7	19.8	0	84.0	9.1	11.2	16.3	310
SE	5.7	7.2	0	0	20.9	25.9	12.1	0	37.6	4.5	4.8	8.1	53
August													
Creel Estimate	28.3	114.5	0	0.6	11.7	115.1	0	4.3	21.9	43.3	50.4	1.2	391
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	28.3	114.5	0	0.6	11.7	115.1	0	4.3	21.9	43.3	50.4	1.2	391
SE	20.3	62.1	0	0.5	5.3	30.0	0	0.0	11.8	37.3	18.5	0.8	84
TOTAL													
Creel Estimate	55	236	0	4	84	259	20	16	150	78	88	17	1,007
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	55	236	0	4	84	259	20	16	150	78	88	17	1,007
SE	21	63	0	2	22	40	12	0	41	38	19	8	117

Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2019. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	21.1	12.7	14.7	0	0	50.4	2.8	5.8	25.0	0	3.7	0	136
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	21.1	12.7	14.7	0	0	50.4	2.8	5.8	25.0	0	3.7	0	136
SE	11.8	1.1	0.8	0	0	14.3	1.4	0.1	7.4	0	1.9	0	20
June													
Creel Estimate	6.2	41.8	0	0.4	21.2	309.2	34.6	0	47.9	107.7	36.9	0	606
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	6.2	41.8	0	0.4	21.2	309.2	34.6	0	47.9	107.7	36.9	0	606
SE	4.4	17.1	0	0.3	7.9	134.8	15.6	0	16.1	63.0	8.1	0	59
July													
Creel Estimate	4.6	0	0	2.7	17.7	207.3	60.6	0	75.0	68.0	14.1	1.6	451
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	4.6	0	0	2.7	17.7	207.3	60.6	0	75.0	68.0	14.1	1.6	451
SE	3.3	0	0	1.6	5.3	37.4	17.6	0	20.8	23.8	6.4	1.3	53
August													
Creel Estimate	113.2	330.9	26.0	3.0	9.4	230.1	5.2	12.0	51.0	61.9	138.5	0	981
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	113.2	330.9	26.0	3.0	9.4	230.1	5.2	12.0	51.0	61.9	138.5	0	981
SE	55.2	143.7	10.9	1.9	3.5	43.8	2.7	0.0	22.7	53.2	45.7	0	177
TOTAL													
Creel Estimate	145	385	41	6	48	797	103	18	199	238	193	2	2,175
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	145	385	41	6	48	797	103	18	199	238	193	2	2,175
SE	57	144	11	3	6	59	18	0	32	58	46	1	240

Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2019. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	10.6	6.4	7.4	4.2	29.5	116.2	41.1	33.3	100.2	13.5	3.7	7.4	373
Lodge Report	18.2	11.0	12.7	0	0	0	0	0	0	0	0	0	42
MAY RETAINED	28.8	17.4	20.1	4.2	29.5	116.2	41.1	33.3	100.2	13.5	3.7	7.4	415
SE	8.4	0.8	0.6	1.8	9.9	31.2	9.6	0.5	28.4	7.6	2.7	5.4	46
June													
Creel Estimate	112.4	181.0	26.4	0	38.9	104.3	52.9	57.9	284.4	193.9	6.6	45.5	1,104
Lodge Report	24.4	81.7	25.8	0	0	0	0	0	0	0	0	0	132
JULY RETAINED	136.8	262.7	52.2	0	38.9	104.3	52.9	57.9	284.4	193.9	6.6	45.5	1,236
SE	35.8	70.3	19.0	0	11.5	26.3	17.8	34.1	40.5	69.6	2.5	40.1	0
July													
Creel Estimate	101.4	213.8	0	0.8	59.0	197.4	73.8	101.9	461.7	276.4	29.5	34.1	1,550
Lodge Report	111.4	180.6	0	0	0	0	0	0	0	0	0	0	292
JULY RETAINED	212.8	394.4	0	0.8	59.0	197.4	73.8	101.9	461.7	276.4	29.5	34.1	1,842
SE	30.9	49.7	0	0.4	17.1	37.2	16.7	31.0	57.4	49.4	6.6	15.4	111
August													
Creel Estimate	268.8	509.0	40.8	0	46.9	199.8	11.6	41.2	187.1	433.0	272.8	9.3	2,020
Lodge Report	0.0	274.0	0.0	0	0	0	0	0	0	0	0	0	274
AUGUST RETAINED	268.8	783.0	40.8	0	46.9	199.8	11.6	41.2	187.1	433.0	272.8	9.3	2,294
SE	82.8	114.1	11.4	0	15.3	37.0	5.2	0.2	33.5	98.7	57.9	6.7	189
TOTAL													
Creel Estimate	493	910	75	5	174	618	179	234	1,033	917	313	96	5,047
Lodge Report	154	547	39	0	0	0	0	0	0	0	0	0	740
TOTAL RETAINED	647	1,458	113	5	174	618	179	234	1,033	917	313	96	5,787
SE	89	124	11	2	25	61	20	31	72	111	58	18	260

Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2019. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	31.7	19.1	22.1	11.6	19.6	50.4	60.9	42.4	112.7	26.9	22.1	36.8	456
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	31.7	19.1	22.1	11.6	19.6	50.4	60.9	42.4	112.7	26.9	22.1	36.8	456
SE	25.1	2.4	1.8	4.7	6.0	14.5	21.7	0.7	31.7	12.7	11.2	26.8	58
June													
Creel Estimate	224.7	320.2	0	0	106.0	186.3	42.7	82.7	467.0	172.4	25.0	27.3	1,654
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	224.7	320.2	0	0	106.0	186.3	42.7	82.7	467.0	172.4	25.0	27.3	1,654
SE	47.7	115.0	0	0	23.2	49.5	10.4	82.7	73.1	46.2	8.5	17.9	130
July													
Creel Estimate	152.1	71.3	8.1	0.4	185.8	345.4	81.7	40.8	776.6	543.7	5.6	16.3	2,228
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	152.1	71.3	8.1	0.4	185.8	345.4	81.7	40.8	776.6	543.7	5.6	16.3	2,228
SE	43.9	27.8	6.7	0.3	42.2	69.6	22.8	29.2	109.6	100.6	2.0	12.7	182
August													
Creel Estimate	537.6	839.9	81.7	0	222.7	569.3	65.7	92.9	347.4	977.3	398.7	0	4,133
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	537.6	839.9	81.7	0	222.7	569.3	65.7	92.9	347.4	977.3	398.7	0	4,133
SE	135.1	301.1	23.0	0	43.5	104.5	26.1	0.5	64.8	247.4	107.8	0	447
TOTAL													
Creel Estimate	946	1,250	112	12	534	1,151	251	259	1,704	1,720	451	80	8,472
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	946	1,250	112	12	534	1,151	251	259	1,704	1,720	451	80	8,472
SE	144	302	24	5	61	126	41	29	131	267	108	30	519

APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month and subarea, in 2019.

Species		Subarea														TOTAL
Month	Adipose	3I	3J	3K	3?	4A	4B	4C	4D	4E	4F	4G	4H	4L	4?	
Chinook Salmon																
May	Clipped	0	0	0	0	0	10	8	2	0	5	0	0	0	0	25
	Not Clipped	2	0	0	0	6	70	50	22	0	22	0	6	0	0	178
	Percent Clipped	0%	-	-	-	0%	13%	14%	8%	-	19%	-	0%	-	-	12%
June	Clipped	0	0	0	0	1	0	2	1	0	9	1	5	0	0	19
	Not Clipped	8	21	9	0	2	53	30	23	0	57	8	31	0	1	243
	Percent Clipped	0%	0%	0%	-	33%	0%	6%	4%	-	14%	11%	14%	-	0%	7%
July	Clipped	1	0	0	0	0	0	0	1	0	6	0	0	0	0	8
	Not Clipped	15	2	1	0	1	11	4	8	2	53	9	8	2	0	116
	Percent Clipped	6%	0%	0%	-	0%	0%	0%	11%	0%	10%	0%	0%	0%	-	6%
August	Clipped	0	0	1	0	0	0	0	1	0	0	0	0	0	0	2
	Not Clipped	2	3	12	0	0	7	7	1	0	11	2	0	0	0	45
	Percent Clipped	0%	0%	8%	-	-	0%	0%	50%	-	0%	0%	-	-	-	4%
Total	Clipped	1	0	1	0	1	10	10	5	0	20	1	5	0	0	54
	Not Clipped	27	26	22	0	9	141	91	54	2	143	19	45	2	1	582
	Percent Clipped	4%	0%	4%	-	10%	7%	10%	8%	0%	12%	5%	10%	0%	0%	8%
Coho Salmon																
May	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Percent Clipped	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
June	Clipped	0	1	0	0	0	0	0	0	0	0	0	1	0	0	2
	Not Clipped	2	7	0	0	0	10	3	3	0	15	0	5	0	1	46
	Percent Clipped	0%	13%	-	-	-	0%	0%	0%	-	0%	-	17%	-	0%	4.2%
July	Clipped	1	0	0	0	0	0	0	0	0	1	0	0	0	0	2
	Not Clipped	28	36	2	0	0	16	9	6	0	78	15	36	10	0	236
	Percent Clipped	3%	0%	0%	-	-	0%	0%	0%	-	1%	0%	0%	0%	-	0.8%
August	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	15	11	59	0	0	58	26	17	0	62	9	1	0	0	258
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	-	0%	0%	0%	-	-	0.0%
Total	Clipped	1	1	0	0	0	0	0	0	0	1	0	1	0	0	4
	Not Clipped	45	54	61	0	0	84	38	26	0	155	24	42	10	1	540
	Percent Clipped	2%	2%	0%	-	-	0%	0%	0%	-	1%	0%	2%	0%	0%	0.7%